Analytics User Guide

Contents

Chapter 1. Audience....4

Chapter 2. Introduction....5

Benefits....5

How does OtoSense ™ work?....5

Access SMS mobile application....6

Chapter 3. Operating Modes....8

Overview....8

Learning Mode....8

Quick Learning....9

Predictive Learning....11

Fully Predictive....11

Normal Operation Mode....11

Motor Condition Mode....12

Access the Motor Attributes' Status....16

Access the Health Index Historical Data....19

Chapter 4. Sensor Information and Specifications....22

Overview....22

Vibration sensors....22

Vibration X-axis Time Waveform....23

Vibration X-axis FFT dB....27

Vibration X-axis FFT Absolute Value....30

Vibration Z-axis Time Waveform....33

Vibration Z-axis FFT dB....37

Vibration Z-axis FFT Absolute Value....40

Magnetic field sensor....43

Magnetic Field Time Waveform....43

Magnetic Field FFT dB....47

Magnetic Field FFT Absolute Value..... 50
Temperature sensors..... 52
Environment Temperature..... 53
Motor Temperature..... 56
Chapter 5. Motor Object..... 60
Overview..... 60
Component Code..... 60
Motor Log Events..... 66
Chapter 6. Appendices..... 68
Appendix A: Miscellaneous API Datasets..... 68
performance..... 68
conditions..... 69
operations..... 71
Appendix B: Cloud Platform Parameters..... 72
Appendix C: ADI OtoSense™ SMS Technical Specifications..... 73
FAQs..... 73
Documents..... 73
Index..... a

Chapter 1. Audience

The analytics user guide is intended for users interested in predictive maintenance analytics.

The analytics user guide is intended for users interested in predictive maintenance analytics.

Chapter 2. Introduction

The ADI OtoSense™ Smart Motor Sensor (SMS) performs predictive maintenance on low-voltage 3-phase induction motors.

The ADI OtoSense™ Smart Motor Sensor (SMS) performs predictive maintenance on low-voltage 3-phase induction motors. It is a complete, easy-to-setup hardware and software solution that uses advanced sensors and AI-based data analysis to:

- provide round-the-clock condition-based monitoring of motor's health. For example, power system, motor shaft balance, and overall performance issues.
- provide fault notifications.
- deliver electrical, mechanical, and thermal health diagnostics with severity levels
- recommend solutions for identified motor and performance issues.

Benefits

This section describes the benefits provided by the SMS.

- This section describes the benefits provided by the SMS.
- The SMS translates the collected data into actionable insights. Here are some of the benefits of using the SMS:
- **Non-intrusive Installation**: The SMS can be attached to active motors, ensuring continuous operation.

  Enhanced Plant Efficiency: The SMS monitors the health of a motor's components and sends alerts about anomalies and degradation that can lead to faults, enabling efficient maintenance planning and reducing unexpected breakdowns.
- **Secure Motor Data**: The SMS safely stores the motor data on the ADI OtoSense™ Server.

  Better Inventory Management: The SMS assists in identifying the required spare components, thereby assisting in minimizing surplus inventory and expenditure.
- **Reduced Maintenance Costs**: The SMS helps extend motor life by providing maintenance insights that reduce repair costs and downtime.

How does OtoSense ™ work?

This section describes how the OtoSense ™ system works.

This section describes how the OtoSense ™ system works.

The hardware and software system consists of the following components:

- The SMS
- Expert health monitoring software hosted on the ADI OtoSense™ Server
- Mobile and web applications that allow customers to monitor the status of motors fitted with an SMS

After an SMS is configured and installed on a motor, it begins sending data to the ADI OtoSense™ server, which then begins a learning process where a unique, customized model is created for that particular motor. It measures the multi-axis vibration, magnetic field, ambient temperature, and temperature condition of the motor. With the help of this data, motor health degradation can be detected that could lead to motor faults in the power system, stator winding, rotor, motor shaft/balance, eccentricity, bearing, alignment, cooling system, attachment (i.e. loose foot), and overall motor performance.

Access SMS mobile application

To access the SMS mobile application, you can use a smartphone or a tablet with

To access the SMS mobile application, you can use a smartphone or a tablet with an Android or iOS operating system.

Note**:** You cannot use a Windows tablet.

The following OS versions are compatible with the SMS mobile app:

- Android: OS6.0 or later
- iOS13 or later

To download the SMS mobile app:

1. Scan the respective QR code from your mobile phone's camera.

Figure a: QR code for iOS devices	Figure b: QR code for Android devices

2. Install the ADI OtoSense™ SMS app from the App Store or from Google Play.
3. Follow the on-screen instructions to complete the installation and open the app.

> **Note:**
> If you have previously signed up on the web application, you can use the same login credentials to sign in to the mobile app.

If this is your first time signing up, follow the instructions in Access to SMS Web Application to sign up on the mobile app.

Figure 1. placeholder

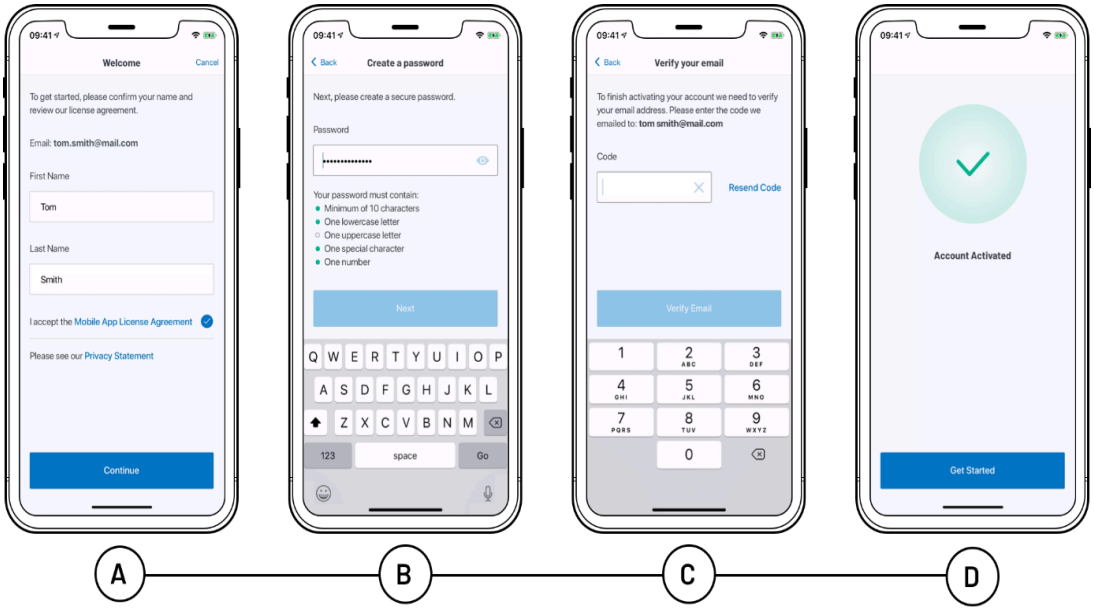


Figure c Welcome pages on SMS mobile app

Chapter 3. Operating Modes

This section describes the operating modes of the OtoSense™ system, which include the learning mode,

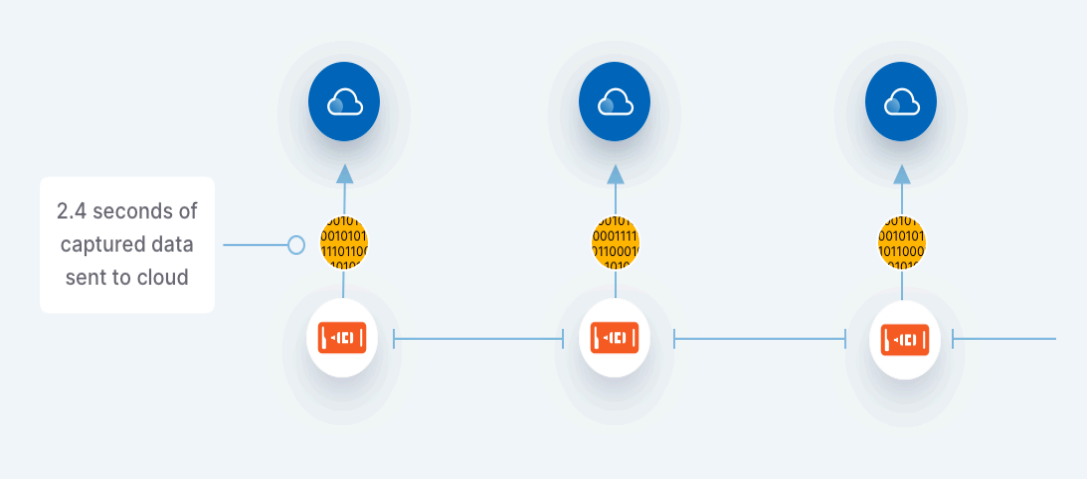
This section describes the operating modes of the OtoSense™ system, which include the learning mode, the normal operation mode, and the motor condition mode.

Overview

After the SMS is configured and installed on a motor, it begins sending data from

After the SMS is configured and installed on a motor, it begins sending data from its on-board sensors to the ADI OtoSense™ Server and initiates a learning process. During this learning process, the analytics create a unique, customized model of the motor whose data is received. At regular intervals, the SMS sends a 2.4 second sample of all sensor data to the ADI OtoSense™ Server.

Figure 2. The SMS Communicating with the Cloud



After the learning mode, the ADI OtoSense™ Server uses the baseline model to detect motor performance anomalies and generate alerts about impending motor issues.

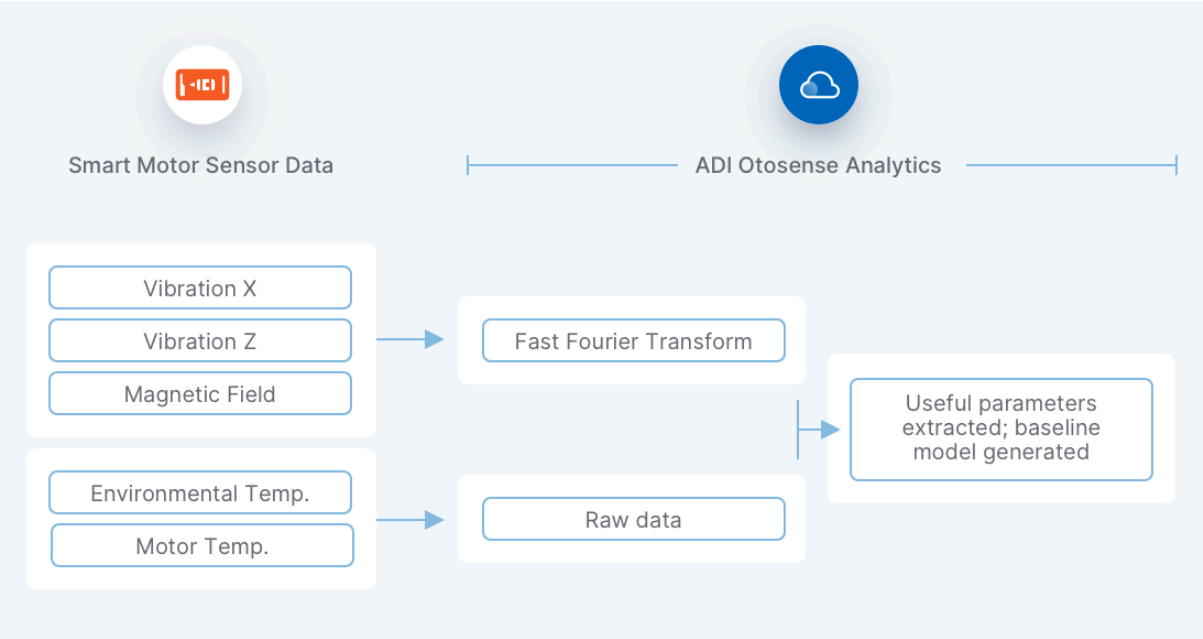
Learning Mode

This section describes the OtoSense™ learning mode.

- This section describes the OtoSense™ learning mode.
- The ADI OtoSense™ SMS requires a learning period to capture the healthy operation of the motor.
- The 2.4 second sample sent by the SMS to the Server includes:

- Vibration X-Axis (raw data)
- Vibration Z-Axis (raw data)
- Magnetic Field (raw data)
- Environmental temperature
- Motor temperature

The proprietary algorithm extracts the most useful parameters from the data samples to generate a baseline model tailored to each motor's operation. This model will be used to monitor changes to the motor's health.



- This learning mode has three stages:

- Quick Learning (approximately seven days)
- Predictive Learning (approximately four weeks)
- Fully Predictive (ongoing)

Note:
Note: During the entire learning mode, the motor should run at all RPMs, all temperatures and with all loads that will be seen by the motor in its normal envelope operation in order to create the best model.

Quick Learning

This section describes the quick learning stage of the learning mode.

This section describes the quick learning stage of the learning mode.

Aseven days are required an initial model of how the motor operates. Faults will not be predicted

During the Quick Learning stage, the status indicator on both the mobile and web application will remain blue with a Learning badge. For more information, see Figure 3 *(on page)*.

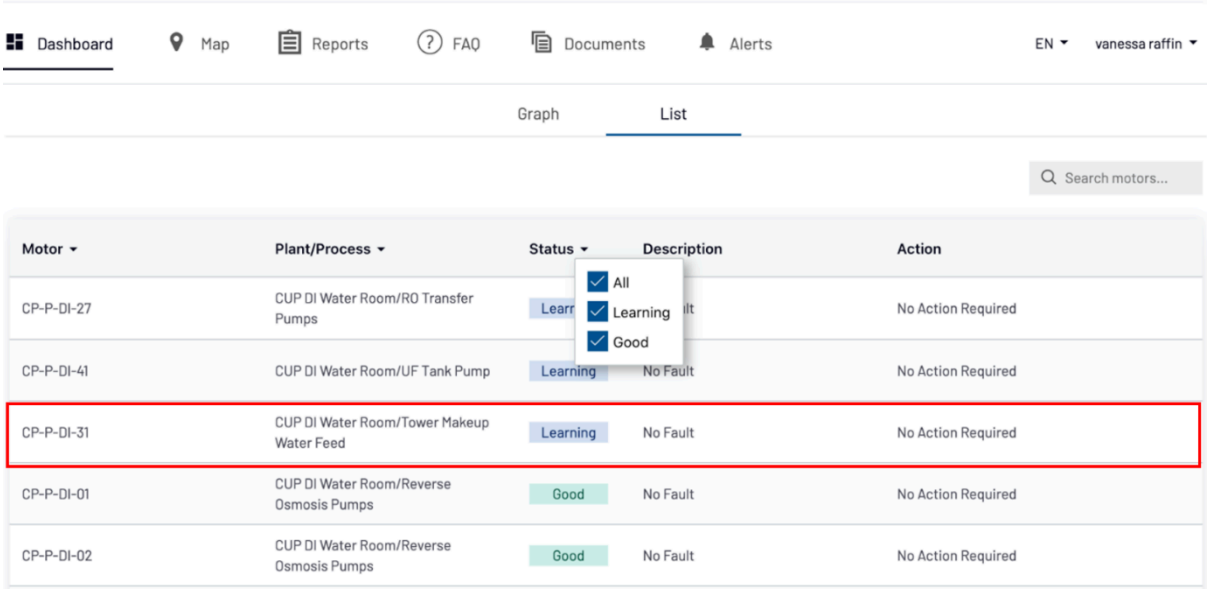
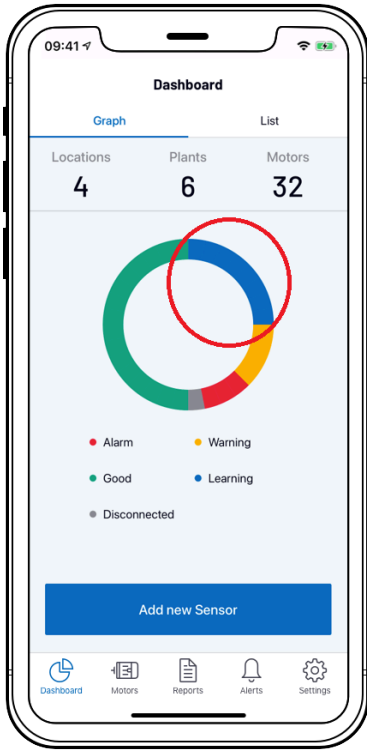


Figure 3. Mobile application screen showing the motors in learning mode



Predictive Learning

This section describes the predictive learning stage of the learning mode.

This section describes the predictive learning stage of the learning mode.

After the Quick Learning stage, OtoSense ™ enters the Predictive Learning stage, which takes approximately four weeks . This stage enables improved predictions for all detectable fault types.

During this stage, faults will be predicted within the parameters of the motor conditions that the SMS observed during Quick Learning. The motor model will continue to improve based on learnings from any changes to operational conditions.

During the Predictive Learning stage, the status indicator on both the mobile and web application will show the motor status. For more information, see Figure 3 *(on page)* and Figure 4 *(on page)*.

Fully Predictive

This section describes the fully predictive stage of the learning mode.

This section describes the fully predictive stage of the learning mode.

When Fully Predictive, the SMS is capable of detecting faults across all conditions observed throughout the Predictive Learning stage.

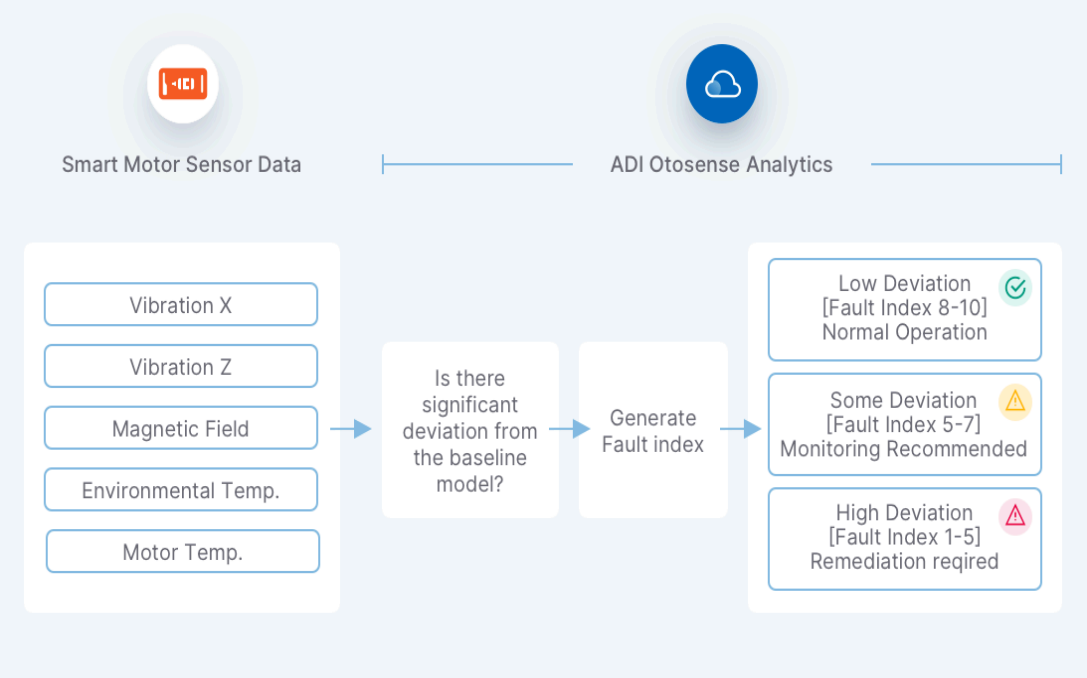
Normal Operation Mode

This section describes the normal operation mode of the SMS.

This section describes the normal operation mode of the SMS.

After the learning mode, the ADI OtoSense™ Cloud platform compares incoming sensor data to the baseline model to detect declining motor health. It identifies significant deviation from the baseline model to generate the health index on a scale of 1-10.

Figure 4. Image8



The REST API fetch the SMS data and display it on the web and mobile applications

- ten key motor attributes; Motor Condition Mode *(on page 12)*
- raw and processed sensor data; Appendix B: Cloud Platform Parameters *(on page)*.

Motor Condition Mode

This section describes the motor condition mode of the SMS.

- This section describes the motor condition mode of the SMS.

  After Normal Operation Mode is established, motor condition mode analyses the deviation from the baseline model to generate the status of the motor. Motor conditioning status is rated on a 1-10 scale.

Table 1. Motor Condition Status Scale

Element	Value	Deviation	Description

Table 1. Motor Condition Status Scale (continued)

Figure 5. placeholder 	8-10	Low Deviation	**normal operation** For example, see Figure 7 *(on page)*.
Figure 6. placeholder 	5-7	Some Deviation	**monitoring is recommended** For example, see Figure 8 *(on page)*.
Figure 7. placeholder 	1-5	High Deviation	**imminent motor failure** For example, see Figure 9 *(on page)*.

Based on incoming sensor data, ADI OtoSense™ Cloud analytics provides insight into nine motor attributes and overall motor performance as displayed in Table 2.

Table 2. Motor Attributes Monitored by the ADI OtoSense™ Cloud Platform

Diagram	Motor Attribute	API Parameter	Type	Description	Example Problem
Figure 8. placeholder 	Power System	`powerSystem`	Electrical	Health of the three phases of the power supply. This could lead to a motor current imbalance.	Lost phase

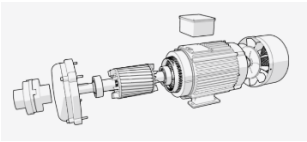
Table 2. Motor Attributes Monitored by the ADI OtoSense™ Cloud Platform (continued)

Figure 9. placeholder 	Stator Winding	`stator Winding`	Electrical	Fault in one of the phases of the motor. This could cause an imbalance of the motor currents.	Coil short circuit
Figure 10. placeholder 	Rotor	`rotor Bar`	Electrical	Fault related to short-circuit ring or rotor bar.	Broken rotor bar
Figure 11. placeholder 	Motor Shaft / Balance	`imbalance`	Electromechanical	Fault related to unequal distribution of mass. This could cause the centre of gravity to shift from the centre of rotation.	Rotor flexion
Figure 12. placeholder 	Eccentricity	`eccentricity`	Mechanical	Fault related to asymmetric air gap between the rotor and the stator.	Bent shaft; improper bearing installation

Table 2. Motor Attributes Monitored by the ADI OtoSense™ Cloud Platform (continued)

Figure 13. placeholder 	Bearing	`bearing`	Mechanical	Fault related to mechanical stress or contamination. This could cause small cracks or defects in the bearing and create vibration problems.	Pitting
Figure 14. placeholder 	Alignment	`misalignment`	Mechanical	Fault that occurs when two rotating shafts (motor and load) are not aligned. This could cause external misalignment.	Angular or parallel misalignment
Figure 15. placeholder 	Cooling System	`cooling`	Temperature	Fault with fans attached to the shaft or externally attached to the motor.	Fan cover collapse
Figure 16. placeholder 	Loose Foot	`motorStructure`	Mechanical	Fault with structural looseness. This occurs when the motor base (or connection	Motor not fixed to base frame

Table 2. Motor Attributes Monitored by the ADI OtoSense™ Cloud Platform (continued)

				to the motor base) is not properly tightened.	
Figure 17. placeholder 	Performance	performance	N/A	Fault due to abnormal Events.	Overall increase in vibrations Unusual changes in motor RPM, motor load or temperatures. This indicates faults that do not correspond to any of the other nine fault categories, which could be related to machine process.

- The ADI OtoSense™ SMS REST API include status and state properties which are fundamental for understanding motor condition mode:
- status- indicates the overall connection status of the motor; online or disconnected, and last online datetime.

  state- indicates the component with the most severe fault i.e. the lowest Fault Score Index (FSI) value. For example, if a motor has multiple components with faults, one an Alarm and another a Warning, the state property of the motor object's status will indicate alarm.
- For more information on the component code and the motor log refer to Motor Object *(on page 60)*.

Access the Motor Attributes' Status

This section describes the process of accessing the motor attributes' status.

This section describes the process of accessing the motor attributes' status.

To access the status of the motor attributes through the web and mobile application:

1. Tap **Motors**.

   The Locations list is displayed.

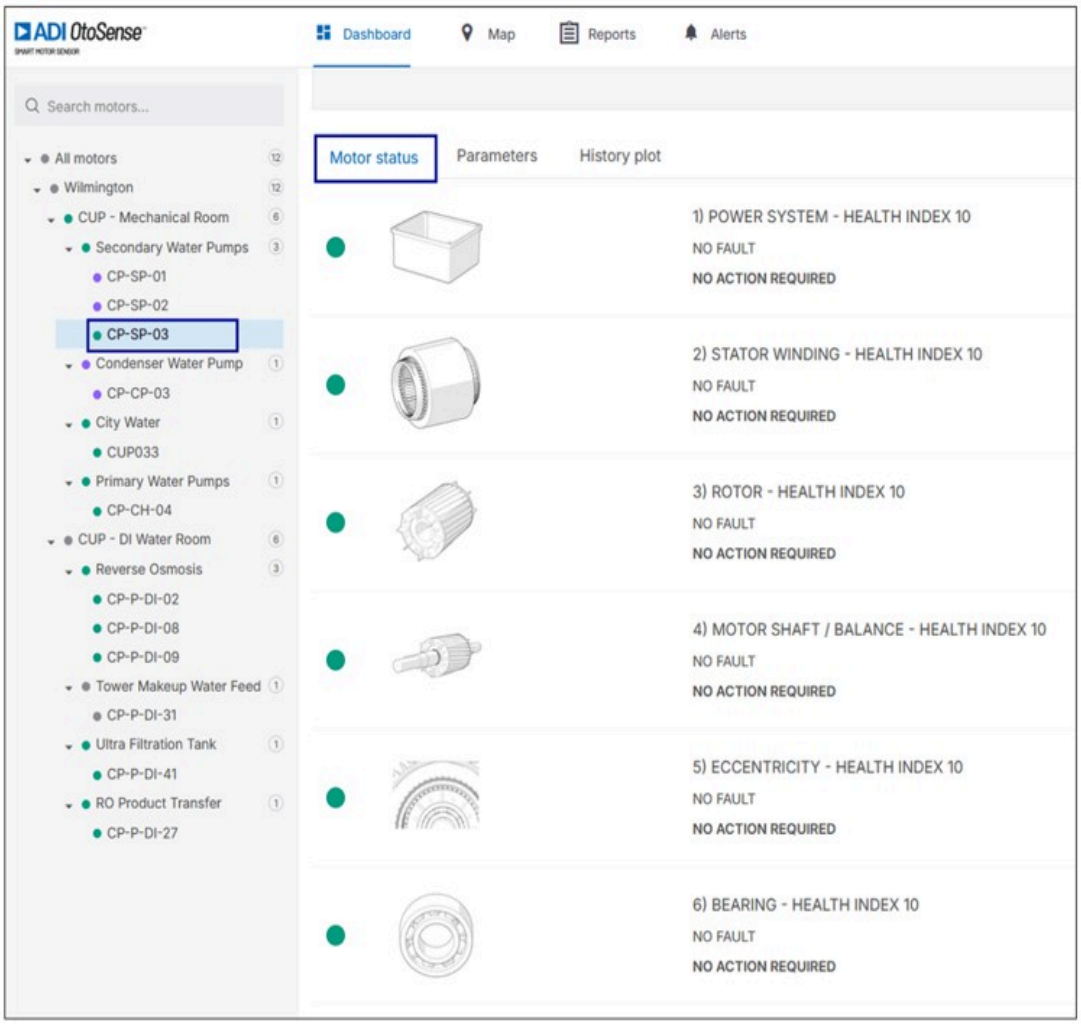
2. Select the location associated with the motor.

3. From the subsequent screens, select the corresponding process, plant, and motor name.

4. Select **Motor s tatus**.

5. The motor condition attributes are displayed in the **Motor status** table as seen below.

   Figure 18. Dashboard screen showing the Motor status



Sample Motor Attributes

This section shows three samples of motor attributes with varying health index values for reference.

This section shows three samples of motor attributes with varying health index values for reference. The motor attribute insights are fetched by the REST API and displayed on the web and mobile application.

The figures display the image and name of the motor attribute, the motor attribute's health index value, the health, and the action required.

Sample 1: Stator Winding Health Index

Stator Winding Health Index 10 on the Dashboard placeholder

Figure 19. Stator Winding Health Index 10 on the Dashboard

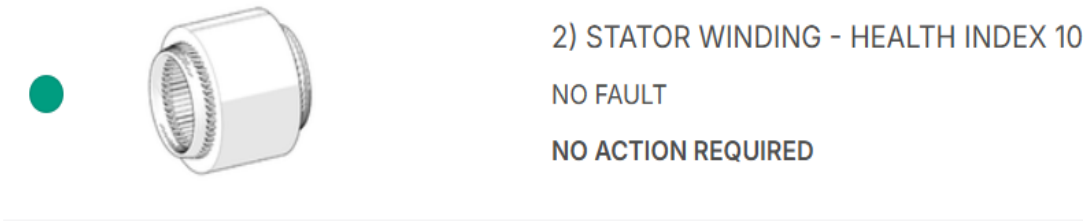


Figure 7 shows that Stator Winding has a value of 10 on the Health Index. Stator Winding is functioning normally. There is no fault and no action is required.

Sample 2: Loose Foot Health Index

Loose Foot Health Index 6 on the Dashboard placeholder

Figure 20. Loose Foot Health Index 6 on the Dashboard

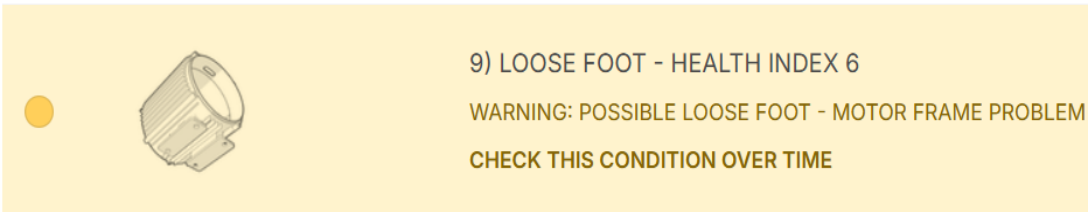


Figure 8 shows that Loose Foot has a value of 6 on the Health Index. Motor frame may not be functioning normally and there is a possible loose foot. User should check the condition over time.

Sample 3: Performance Health Index

Performance Health Index 3 and Suggested Remediation Action on the Dashboard placeholder

Figure 21. Performance Health Index 3 and Suggested Remediation Action on the Dashboard

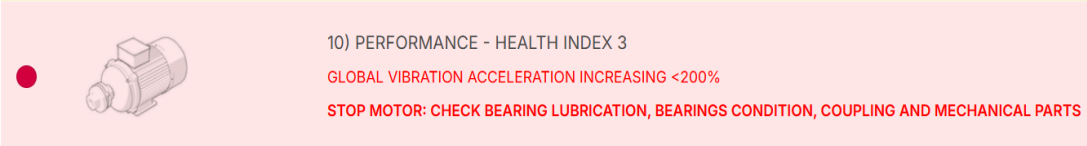


Figure 9 shows that Performance has a value of 3 on the Health Index. Global Vibration Acceleration is increasing up to 200%. User should stop the motor immediately and check the bearing lubrication, bearing condition, coupling, and mechanical parts.

Access the Health Index Historical Data

This section describes the process of accessing the health index historical data.

This section describes the process of accessing the health index historical data.

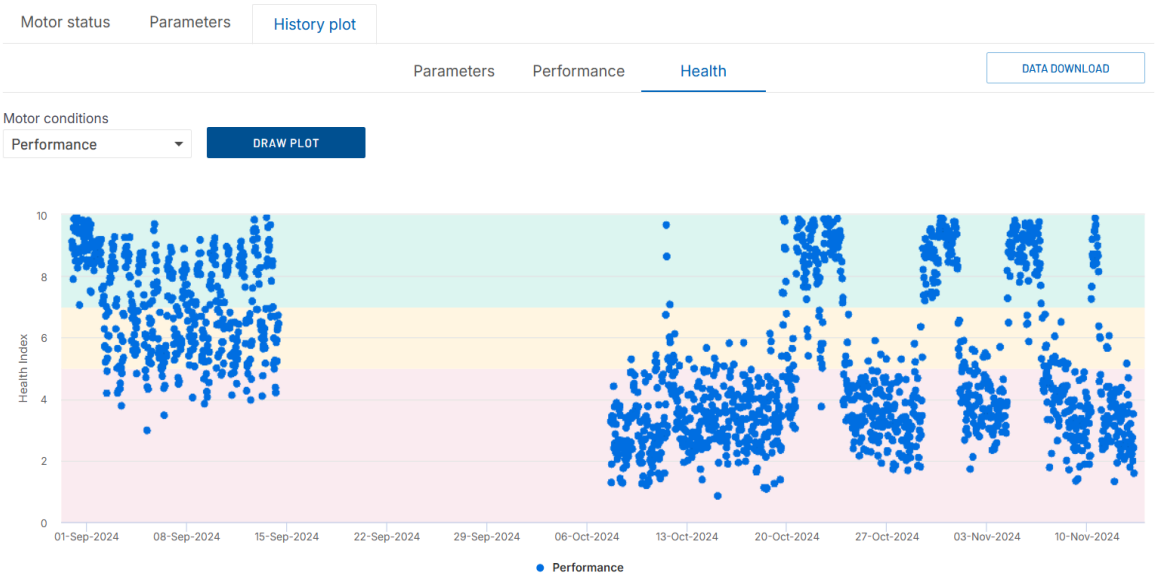
To access the Health Index Historical Data through the web and mobile application:

1. Tap **Motors**.

   The Locations list is displayed.

2. Select the location associated with the motor.
3. From the subsequent screens, select the corresponding process, plant, and motor name.
4. Select **History plot**and**Health**.
5. Tap **Select …** to select **Performance**.
6. Click **DRAW PLOT**.

   You should see the corresponding scatter plot as shown in Figure 10 *(on page)*.



The scatter plot has a legend at the bottom with element and its corresponding motor attribute.

7. Click **DATA DOWNLOAD**to download the generated data.

Sample Health Index Historical Data

Figure 10 shows the Health Index Historical Data as a scatter plot generated by analysing

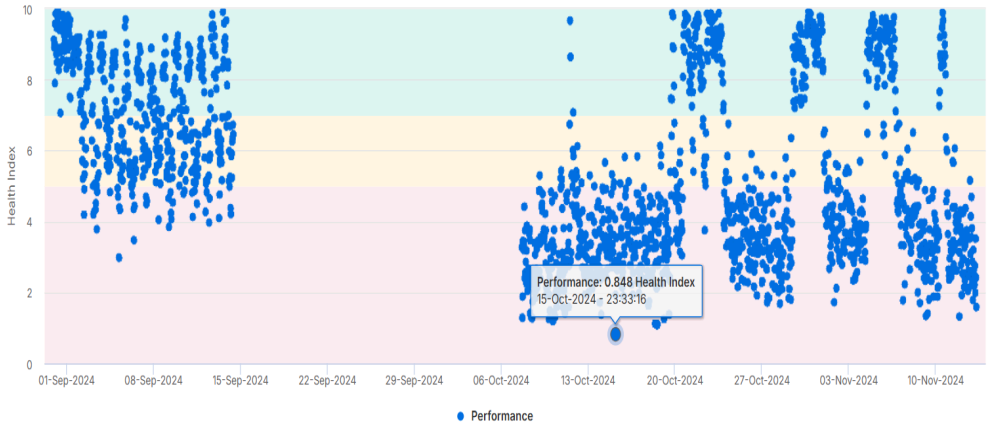
Figure 10 shows the Health Index Historical Data as a scatter plot generated by analysing the data received from the SMS by the ADI OtoSense™ Server.

- The scatter plot's **X-axis** shows the 1-10 scale of the Health Index.
- The scatter plot's **Y-axis** shows all dates between 01 September 2024 and 10 November 2024.
- The

  icon is a data point for the Performance parameter of the motor.
- You can hover on each data point (

) to view the health index value, date, and time.

- Figure 22. Image28



- You can analyse the clusters and outliers to better understand the underlying issues.

  For example, data points that fall below 5 on the health index should be further analysed and appropriate action should be taken.

Chapter 4. Sensor Information and Specifications

This section describes the information and specifications related to the sensors on the SMS.

This section describes the information and specifications related to the sensors on the SMS.

Overview

The ADI OtoSense™ SMS has three on-board sensors that collect the data used to monitor

The ADI OtoSense™ SMS has three on-board sensors that collect the data used to monitor any significant deviation from the established baseline model:

- **Vibration sensors** – sense vibration in the X- and Z- axes.
- **Magnetic field sensor** - measures the magnetic field around a motor.
- **Temperature sensors** – measure temperatures of the motor and the environment.

The REST API fetch data to display status updates on the web and mobile application. For more information, see Appendix B: Cloud Platform Parameters *(on page)*.

Vibration sensors

This section describes the information and specifications related to vibration sensors on the SMS.

- This section describes the information and specifications related to vibration sensors on the SMS.

  Vibrations monitoring and analysis are at the core of most condition-based monitoring systems as vibrations and harmonic signatures are often the first indication of an impending motor failure.

  ADI OtoSense™ SMS is equipped with a Vibration X-axis (tangential vibration) sensor and a Vibration Z-axis (axial vibration) sensor. For specifications, see Table 3.

Table 3. Vibration Sensor Specifications

Sensor Type	Amplitude Range	Frequency Range	Sampling Frequency	Number of Samples	Data Formats	Data Type
X-Axis Vibration						
Y-Axis Vibration						

Since each motor has a unique vibrational profile, the ADI OtoSense™ SMS first determines a motor's baseline operating condition during the Learning Mode.

The REST API fetch the time and frequency domain data to display it on the web and mobile application. Signal processing allows for the quick identification of key signal characteristics which form the basis of the analysis.

Vibration X-axis Time Waveform

This section describes the Vibration X-axis time waveform. It is generated from data received by

This section describes the Vibration X-axis time waveform. It is generated from data received by the ADI OtoSense™ Server from the vibration X-axis sensor on the SMS.

Access the Vibration X-axis Time Waveform

This section describes the process of accessing the Vibration X-axis time waveform graph.

This section describes the process of accessing the Vibration X-axis time waveform graph.

To access the Vibration X-axis Time Waveform through the application:

1. Tap **Motors**.

   The Locations list is displayed.

2. Select the location associated with the motor.

3. From the subsequent screens, select the corresponding process, plant, and motor name.

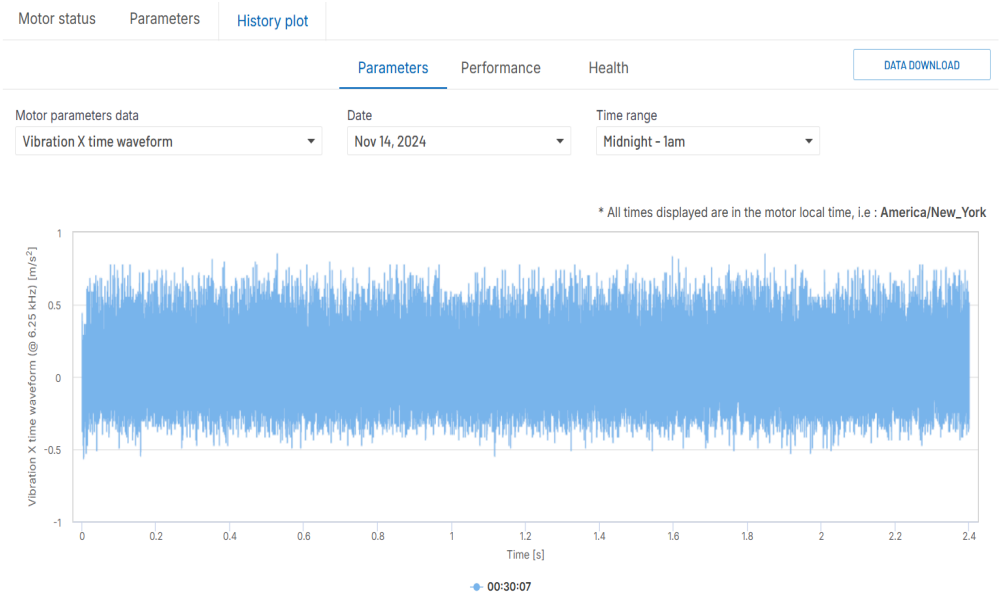
4. Select **History Plot** and **Parameters**.

5. Tap **Select Vibration** to select **Vibration X time waveform**.

6. Tap **Select Date** and **Select Time** to select the date and time corresponding to the moment of the installation.

   You should see the corresponding waveform as shown in Figure 12 *(on page)*.

Figure 23. Dashboard screen showing the Vibration X time waveform



7. Click **DATA DOWNLOAD**to download the generated data.

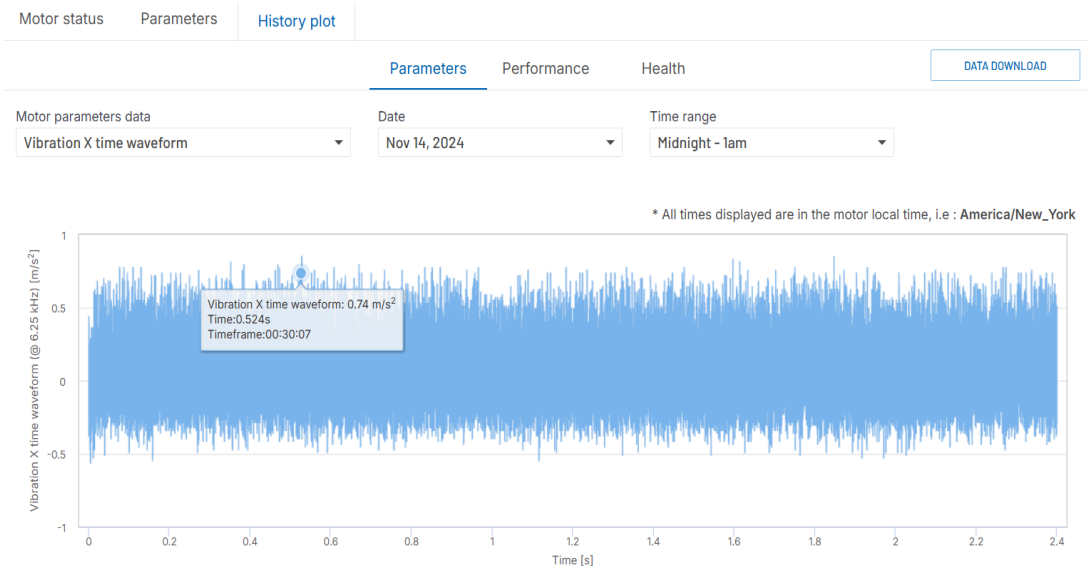
Data Visualization

This section describes the Vibration X-axis time waveform graph.

This section describes the Vibration X-axis time waveform graph.

You can hover on a section of the Vibration X time waveform graph to view the Vibration X time waveform in metre per second squared (m/s2), the time in seconds, and the time frame.

Figure 24. Dashboard screen highlighting one data point of the Vibration X time waveform



Sample Data

Three overlaid samples of the Vibration X time waveform placeholder

Figure 25. Three overlaid samples of the Vibration X time waveform

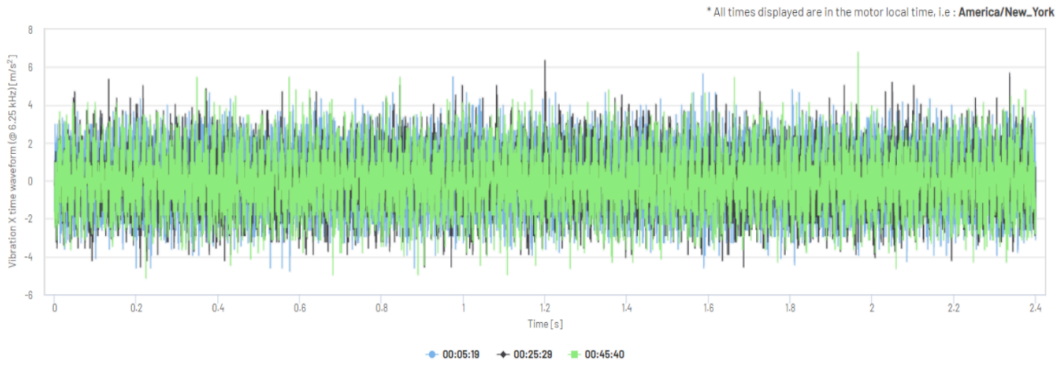


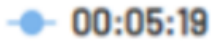
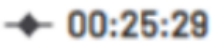
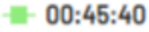
Figure 14 is a waveform graph showing three overlaid waveform samples from the Vibration X-axis sensor.

- The graph's X-axis shows the time in seconds.
- The graph's Y-axis shows the vibration time waveform in metre per second squared (m/s2) at 6.25 kHz sample frequency.

Table 4. Vibration X time waveform Legend

Legend	Description

Table 4. Vibration X time waveform Legend (continued)

Figure 26. placeholder 	Waveform data at 5 minutes 19 seconds
Figure 27. placeholder 	Waveform data at 25 minutes 29 seconds
Figure 28. placeholder 	Waveform data at 45 minutes 40 seconds

API Dataset

This section describes the API dataset used for the vibration X-axis time waveform graph.

This section describes the API dataset used for the vibration X-axis time waveform graph.

The REST APIs have access to the following dataset:

vibx

The Vibration X (vibx) is the vibration acceleration measured in the tangential direction of the

The Vibration X (vibx) is the vibration acceleration measured in the tangential direction of the motor. This recorded data shows the vibration X waveform data, in the unit of metre per second squared (m/s2).

- Each record contains an array of 15,000 samples captured on the specified timestamp.
- The X-axis data is a vector of 15,000 data points with a step of 0.00016s.
- For example:
- [0, 0.00016, 0.00032, 0.00048, ... 2.39984]

The following two datasets, vibration Z (vibz) and magnetic field (flux), also use the same time stamp to generate the waveform graphic. All 3 datasets, vibx, vibz

- For more information on API refer to the ADI OtoSense ™ SMS REST API.

Vibration X-axis FFT dB

This section describes the Vibration X-axis FFT. It is generated from data received by the

This section describes the Vibration X-axis FFT. It is generated from data received by the ADI OtoSense™ Server from the vibration X-axis sensor on the SMS.

Access the Vibration X-axis FFT dB

This section describes the process of accessing the Vibration X-axis FFT graph.

This section describes the process of accessing the Vibration X-axis FFT graph.

To access the Vibration X-axis FFT dB through the application:

1. Tap **Motors**.

   The Locations list is displayed.

2. Select the location associated with the motor.

3. From the subsequent screens, select the corresponding process, plant, and motor name.

4. Select **History Plot** and **Parameters**.

5. Tap **Select Vibration** to select **Vibration X FFT dB**.

6. Tap **Select Date** and **Select Time** to select the date and time corresponding to the moment of the installation.

   You should see the corresponding waveform as shown in Figure 15 *(on page)*.

Figure 29. Dashboard screen showing the Vibration X FFT waveform



7. Click **DATA DOWNLOAD**to download the generated data.

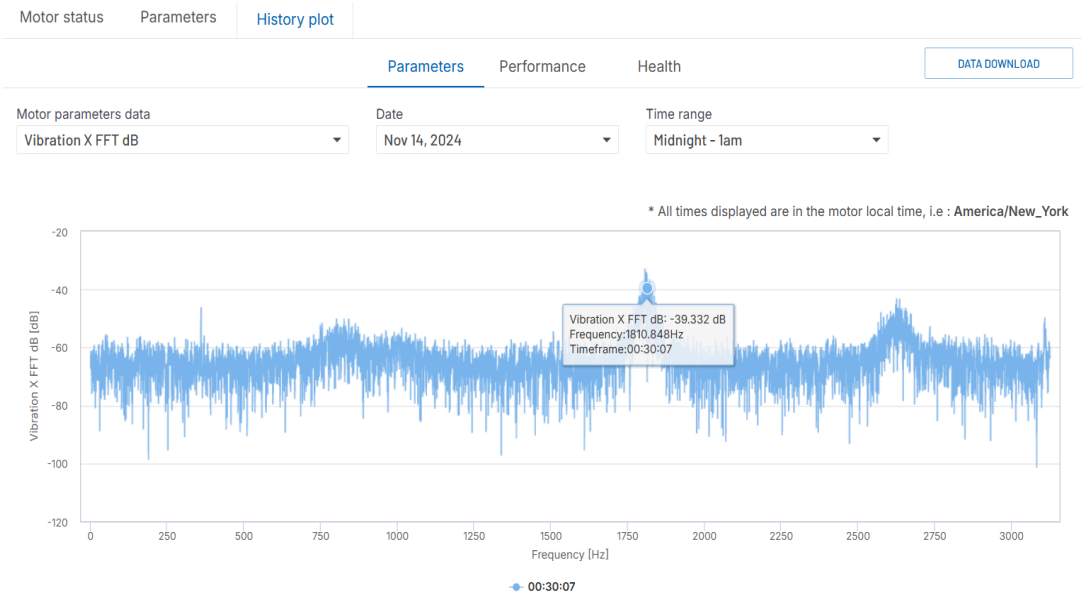
Data Visualization

This section describes the Vibration X-axis FFT graph.

This section describes the Vibration X-axis FFT graph.

You can hover on a section of the Vibration X FFT dB graph to view the Vibration X FFT in decibels (dB), the frequency in Hertz (Hz), and the time frame.

Figure 30. Dashboard screen highlighting one data point of the Vibration X FFT waveform



Sample Data

Sample of the Vibration X FFT waveform placeholder

Figure 31. Sample of the Vibration X FFT waveform



Figure 17 is a waveform graph showing Fast Fourier Transform in decibels data from the X-axis sensor.

- The graph's X-axis shows frequency in Hertz.
- The graph's Y-Axis shows the FFT of the X-Axis accelerometer signal in decibels.

API Dataset

This section describes the API dataset used for the vibration X-axis FFT graph.

This section describes the API dataset used for the vibration X-axis FFT graph.

The REST APIs have access to the following dataset:

vibxFFT

The Vibration X FFT (vibxFFT) is the vibration amplitude measured in the tangential direction of

The Vibration X FFT (vibxFFT) is the vibration amplitude measured in the tangential direction of the motor. This recorded data is the result of the FFT transformation with the vibration signal X, in the unit of decibels (dB).

- Each record contains an FFT of the vibx samples captured on the specified timestamp.

  It is necessary to generate the corresponding X-axis data to plot this vibxFFT data in a graphic. The X-axis data is a vector of 7,500 data points with a step of 0.41667 Hertz (Hz).

- For example:

- [0.41667, 0.83334, 1.25, ... , 3125]

  The following two datasets, Vibration Z FFT (vibzFFT) and magnetic field FFT (fluxFFT), also use the same time stamp to generate the FFT graphic. All 3 datasets, vibxFFT, vibzFFT

- For more information on API refer to the ADI OtoSense™ SMS REST API.

Vibration X-axis FFT Absolute Value

This section describes the Vibration X-axis FFT Absolute Value. It is generated from data received

This section describes the Vibration X-axis FFT Absolute Value. It is generated from data received by the ADI OtoSense™ Server from the vibration X-axis sensor on the SMS.

Access the Vibration X-axis FFT Absolute Value

This section describes the process of accessing the Vibration X-axis FFT Absolute Value graph.

This section describes the process of accessing the Vibration X-axis FFT Absolute Value graph.

To access the Vibration X-axis FFT Absolute Value through the application:

1. Tap **Motors**.

   The Locations list is displayed.

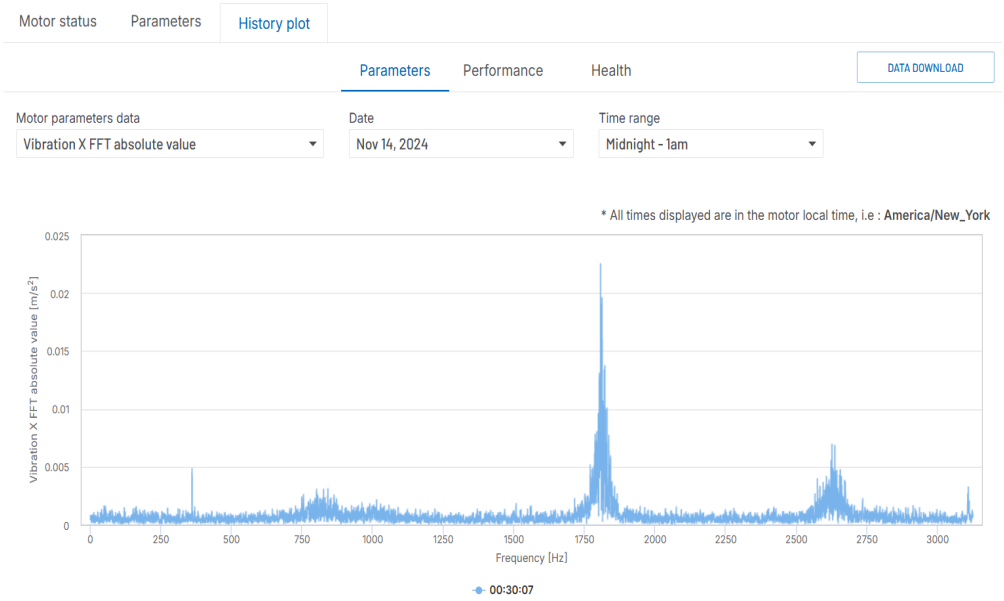
2. Select the location associated with the motor.

3. From the subsequent screens, select the corresponding process, plant, and motor name.

4. Select **History Plot** and **Parameters**.
5. Tap **Select Vibration** to select **Vibration X absolute value**.
6. Tap **Select Date** and **Select Time** to select the date and time corresponding to the moment of the installation.

   You should see the corresponding waveform as shown in Figure 18 *(on page)*.

Figure 32. Dashboard screen showing the Vibration X FFT absolute value waveform



7. Click **DATA DOWNLOAD**to download the generated data.

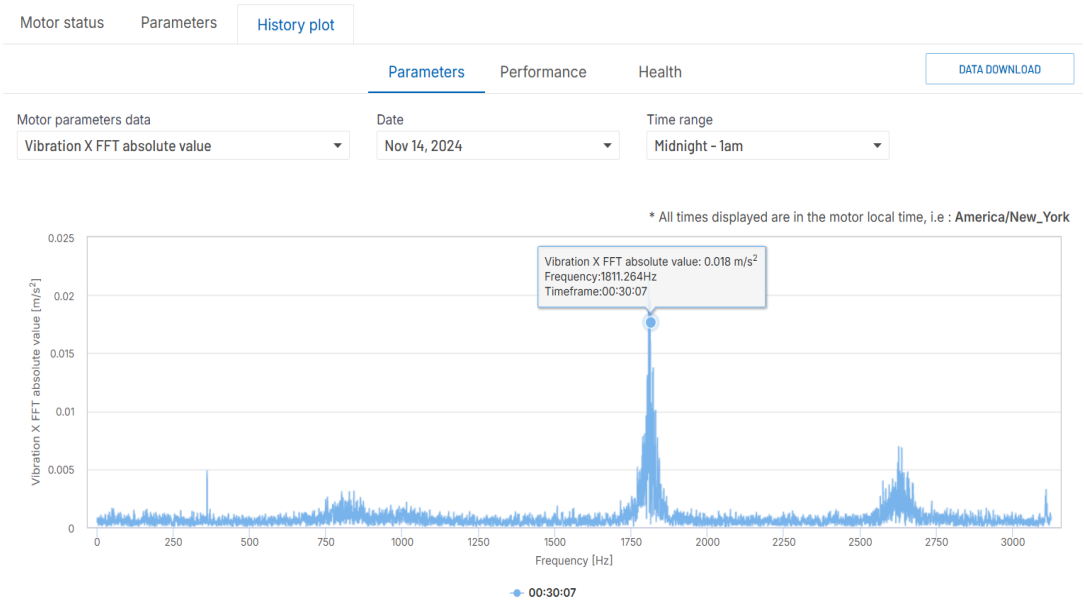
Data Visualization

This section describes the Vibration X-axis FFT Absolute Value graph.

This section describes the Vibration X-axis FFT Absolute Value graph.

You can hover on a section of the Vibration X FFT absolute value graph to view the Vibration X FFT absolute value in metre per second squared (m/s2), the frequency in Hertz (Hz), and the time frame.

Figure 33. Dashboard screen highlighting one data point of the Vibration X FFT absolute value waveform



Sample Data

Sample of the Vibration X FFT absolute value waveform placeholder

Figure 34. Sample of the Vibration X FFT absolute value waveform

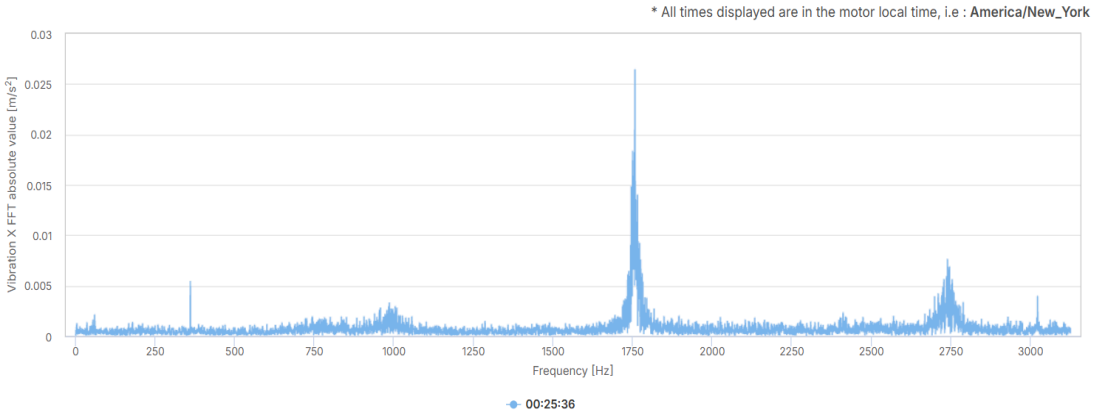


Figure 20 is a waveform graph showing Fast Fourier Transform absolute value in metres per second squared (m/s2) data from the X-axis sensor.

- The graph's X-axis shows frequency in Hertz.
- The graph's Y-Axis shows the absolute value of the FFT of the X-Axis accelerometer signal in metres per second squared (m/s2).

Vibration Z-axis Time Waveform

This section describes the Vibration Z-axis time waveform. It is generated from data received by

This section describes the Vibration Z-axis time waveform. It is generated from data received by the ADI OtoSense™ Server from the vibration Z-axis sensor on the SMS.

Access the Vibration Z-axis Time Waveform

This section describes the process of accessing the Vibration Z-axis time waveform graph.

This section describes the process of accessing the Vibration Z-axis time waveform graph.

To access the Vibration Z-axis Time Waveform through the application:

1. Tap **Motors**.

   The Locations list is displayed.

2. Select the location associated with the motor.

3. From the subsequent screens, select the corresponding process, plant, and motor name.

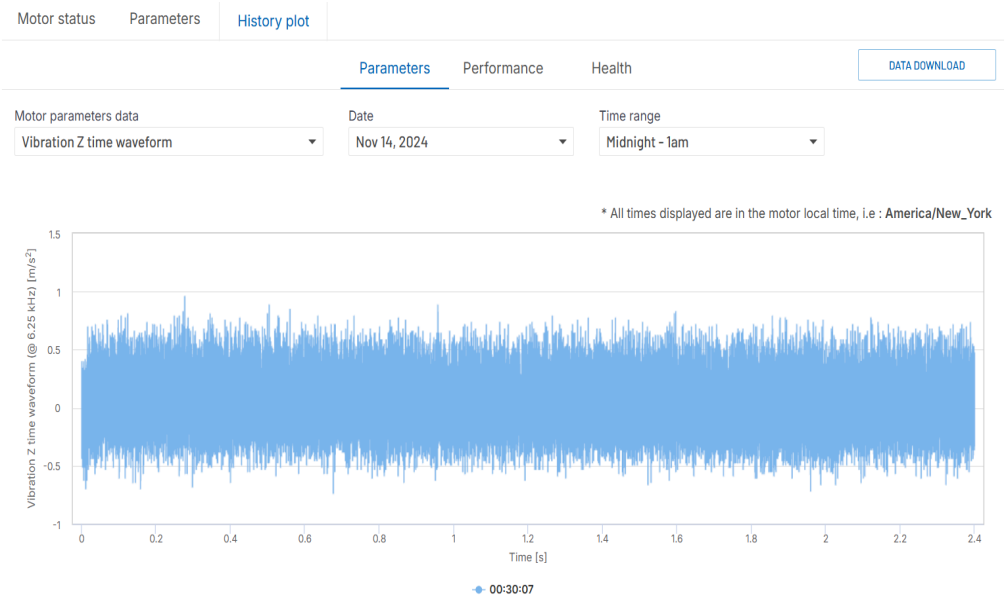
4. Select **History Plot** and **Parameters**.

5. Tap **Select Vibration** to select **Vibration Z time waveform**.

6. Tap **Select Date** and **Select Time** to select the date and time corresponding to the moment of the installation.

   You should see the corresponding waveform as shown in Figure 21 *(on page)*.

Figure 35. Dashboard screen showing the Vibration Z time waveform



7. Click **DATA DOWNLOAD**to download the generated data.

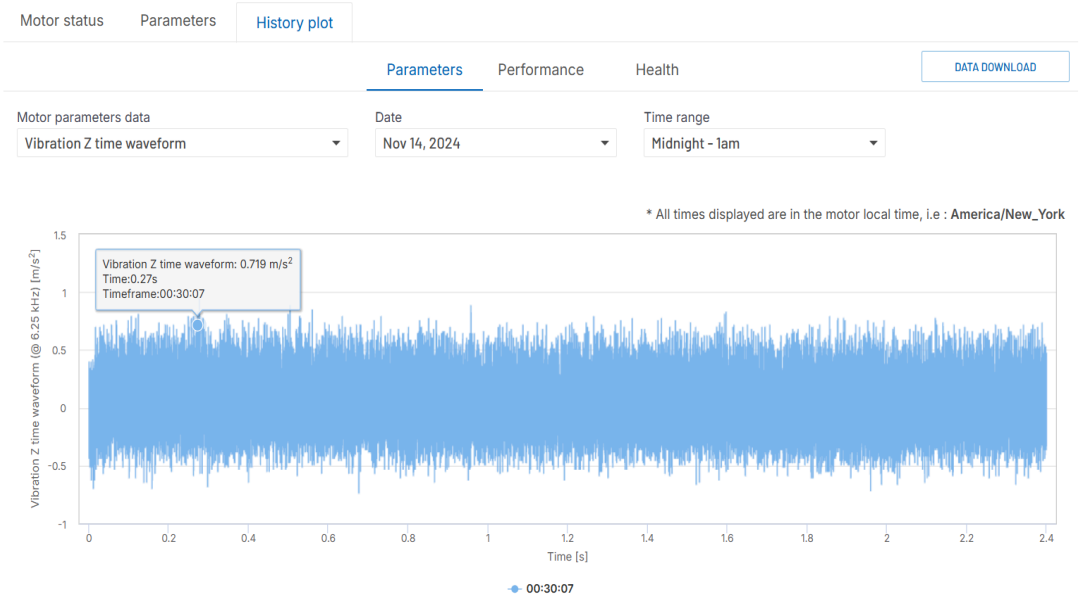
Data Visualization

This section describes the Vibration Z-axis time waveform graph.

This section describes the Vibration Z-axis time waveform graph.

You can hover on a section of the Vibration Z time waveform graph to view the Vibration Z waveform value in metre per second squared (m/s2), the time in seconds, and the time frame.

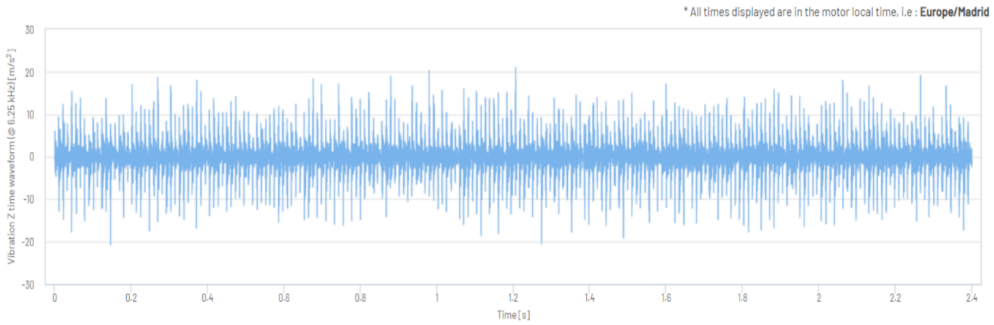
Figure 36. Dashboard screen highlighting one data point of the Vibration Z time waveform



Sample Data

2.4 seconds sample of the Vibration Z time waveform placeholder

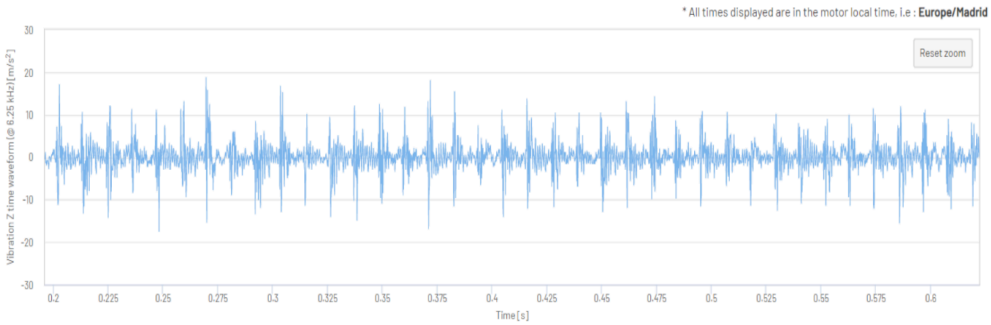
- Figure 37. 2.4 seconds sample of the Vibration Z time waveform



- Figure 23 is a waveform graph showing time waveform data from the Z-axis sensor.

- The graph's X-axis shows the time in seconds.
- The graph's Y-Axis shows the time waveform in metre per second squared (m/s2) at the 6.25 kHz sample frequency.

- Figure 38. Zoomed-in view of the 2.4 seconds sample of Vibration Z time waveform between 0.95 seconds and 1.45 seconds



- Figure 25: Zoomed-in view of the 2.4 seconds sample of Vibration Z time waveform between 0.2 seconds and 0.6 seconds

  Figure 24 and 25 show a zoomed-in view of the waveform graph showing Fast Fourier Transform data from the Z-axis sensor.

- The graph's X-axis shows the time in seconds.
- The graph's Y-Axis shows the time waveform in metre per second squared (m/s2) at 6.2 KHz sample frequency.

API Dataset

This section describes the API dataset used for the vibration Z-axis time waveform graph.

This section describes the API dataset used for the vibration Z-axis time waveform graph.

The REST APIs have access to the following dataset:

vibz

The Vibration Z (vibz) is the vibration acceleration measured in the axial direction of the

The Vibration Z (vibz) is the vibration acceleration measured in the axial direction of the motor. This recorded data shows the vibration Z waveform data, in the unit of metre per second squared (m/s2). With the vibrations X and Z measured from the motor, the analytics system can detect the mechanical failures of the motor.

- Each record contains an array of 15,000 samples captured on the specified timestamp.

  It is necessary to generate the corresponding X-axis data to plot this Vibration Z (vibx) waveform data in a graphic: The X-axis data is a vector of 15,000 data points with a step of 0.00016s.

- For example:
- [0, 0.00016, 0.00032, 0.00048, ... 2.39984]
- For more information on API refer to the ADI OtoSense™ SMS REST API.

Vibration Z-axis FFT dB

This section describes the Vibration Z-axis FFT. It is generated from data received by the

This section describes the Vibration Z-axis FFT. It is generated from data received by the ADI OtoSense™ Server from the vibration Z-axis sensor on the SMS.

Access the Vibration Z-axis FFT dB

This section describes the process of accessing the Vibration Z-axis FFT graph.

This section describes the process of accessing the Vibration Z-axis FFT graph.

To access the Vibration Z-axis FFT dB through the application:

1. Tap **Motors**.

   The Locations list is displayed.

2. Select the location associated with the motor.
3. From the subsequent screens, select the corresponding process, plant, and motor name.
4. Select **History Plot** and **Parameters**.
5. Tap **Select Vibration** to select **Vibration Z FFT dB**.
6. Tap **Select Date** and **Select Time** to select the date and time corresponding to the moment of the installation.

   You should see the corresponding waveform as shown in Figure 26 *(on page)*.

Figure 39. Dashboard screen showing the Vibration Z FFT waveform



7. Click **DATA DOWNLOAD**to download the generated data.

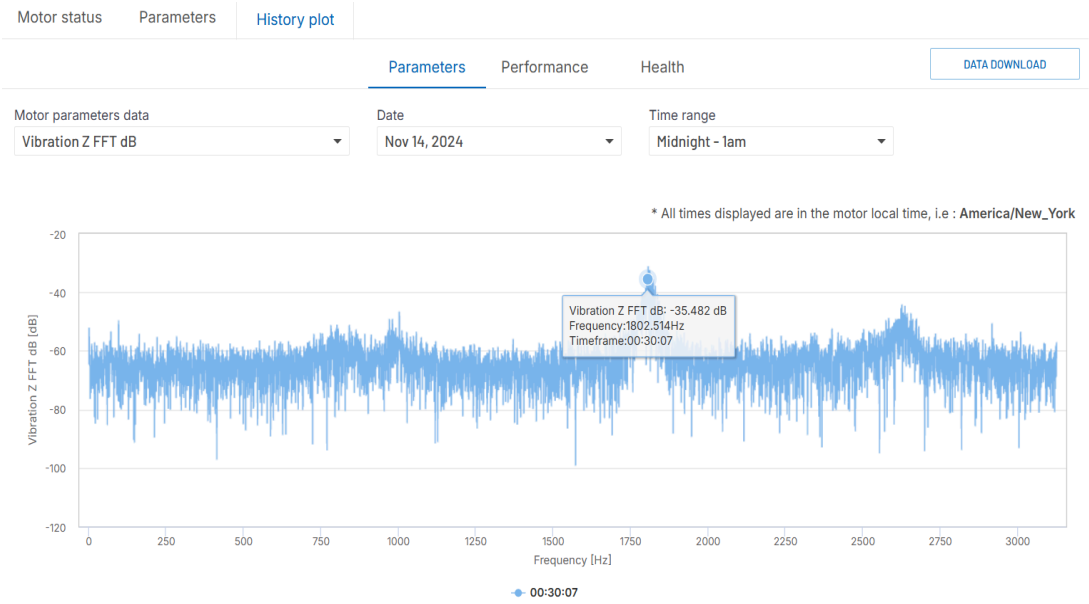
Data Visualization

This section describes the Vibration Z-axis FFT graph.

This section describes the Vibration Z-axis FFT graph.

You can hover on a section of the Vibration Z FFT dB graph to view the Vibration Z FFT dB value in decibels (dB), the frequency in Hertz (Hz), and the time frame as shown in Figure 27 *(on page)*.

Figure 40. Dashboard screen highlighting one data point of the Vibration Z FFT waveform



Sample Data

2.4 seconds sample of the Vibration Z FFT waveform placeholder

Figure 41. 2.4 seconds sample of the Vibration Z FFT waveform



Figure 28 shows the 2.4 second sample of the Fast Fourier Transform data from the Z-axis sensor at 5 minutes 19 seconds (

).

- The graph's X-axis shows the frequency in Hertz.
- The graph's Y-Axis shows the Fast Fourier Transform data in decibels.

API Dataset

This section describes the API dataset used for the vibration Z-axis FFT graph.

This section describes the API dataset used for the vibration Z-axis FFT graph.

The REST APIs have access to the following dataset:

vibzFFT

The Vibration Z FFT (vibzFFT) is the vibration amplitude measured in the axial direction of

The Vibration Z FFT (vibzFFT) is the vibration amplitude measured in the axial direction of the motor. The recorded data is the result of the FFT transformation of the vibration signal Z, in the unit of decibels (dB).

- Each record contains an FFT of the vibz samples captured on the specified timestamp.

  It is necessary to generate the corresponding X-axis data to plot this vibzFFT data in a graphic. The X-axis data is a vector of 7,500 data points with a step of 0.41667 Hertz (Hz).

- For example:

- [0.41667, 0.83334, 1.25, ... , 3125]

- For more information on API refer to the ADI OtoSense™ SMS REST API.

Vibration Z-axis FFT Absolute Value

This section describes the Vibration Z-axis FFT Absolute Value. It is generated from data received

This section describes the Vibration Z-axis FFT Absolute Value. It is generated from data received by the ADI OtoSense™ Server from the vibration Z-axis sensor on the SMS.

Access Vibration Z-axis FFT Absolute Value

This section describes the process of accessing the Vibration Z-axis FFT Absolute Value graph.

This section describes the process of accessing the Vibration Z-axis FFT Absolute Value graph.

To access the Vibration Z-axis FFT Absolute Value through the application:

1. Tap **Motors**.

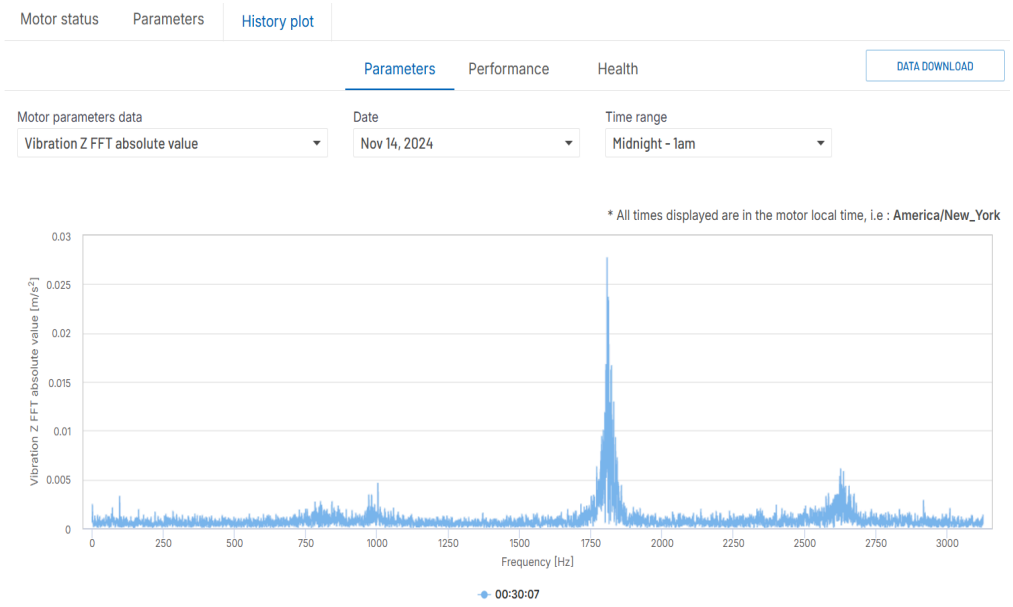
   The Locations list is displayed.

2. Select the location associated with the motor.

3. From the subsequent screens, select the corresponding process, plant, and motor name.
4. Select **History Plot** and **Parameters**.
5. Tap **Select Vibration** to select **Vibration Z FFT absolute value**.
6. Tap **Select Date** and **Select Time** to select the date and time corresponding to the moment of the installation.

   You should see the corresponding waveform as shown in Figure 28 *(on page)*.

Figure 42. Dashboard screen showing the Vibration Z FFT absolute value waveform



7. Click **DATA DOWNLOAD**to download the generated data.

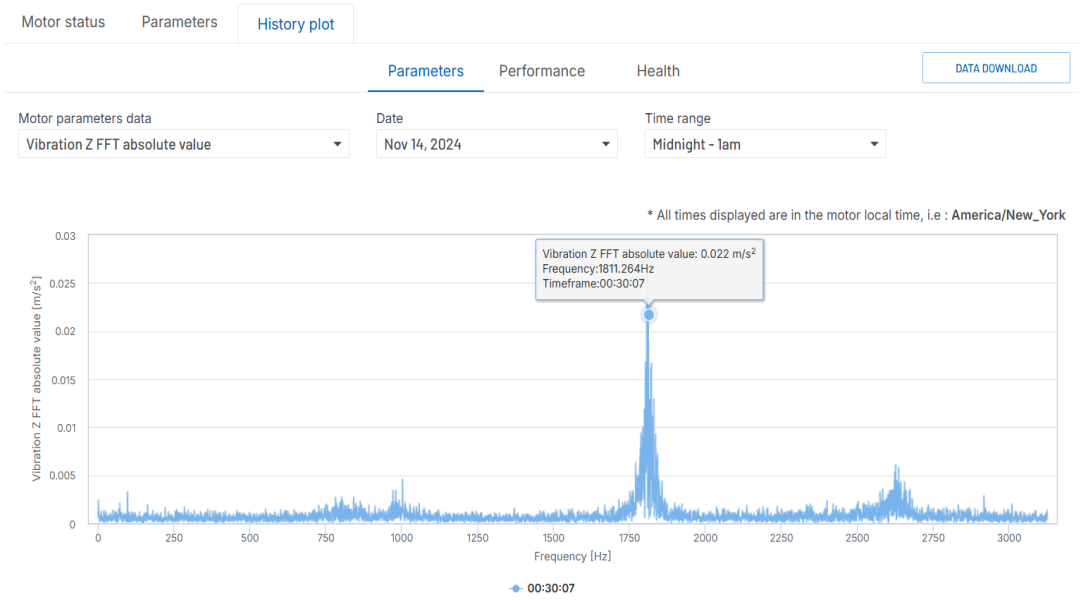
Data Visualization

This section describes the Vibration Z-axis FFT Absolute Value graph.

This section describes the Vibration Z-axis FFT Absolute Value graph.

You can hover on a section of the Vibration Z FFT absolute value graph to view the Vibration Z FFT absolute value in metre per second squared (m/s2), the frequency in Hertz (Hz), and the time frame.

Figure 43. Dashboard screen highlighting one data point of the Vibration Z FFT absolute value waveform



Sample Data

Sample of the Vibration X FFT absolute value waveform placeholder

Figure 44. Sample of the Vibration X FFT absolute value waveform

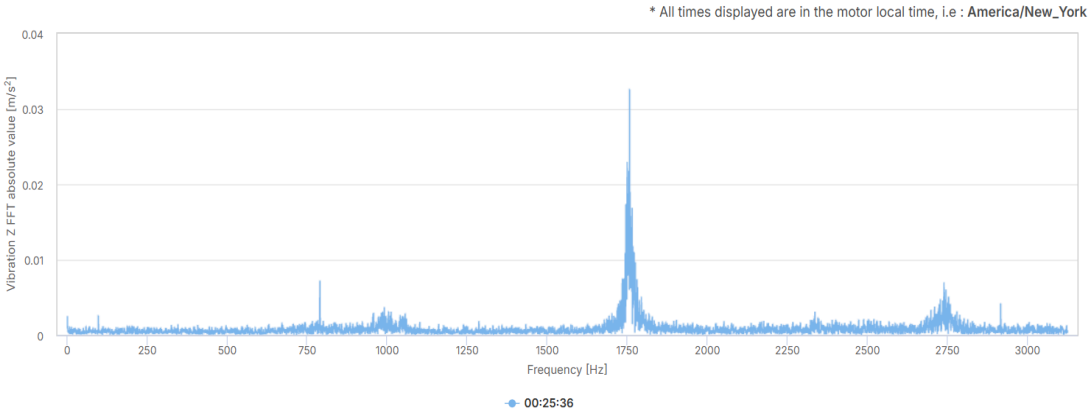


Figure 30 is a waveform graph showing Fast Fourier Transform absolute value in metres per second squared (m/s2) data from the Z-axis sensor.

- The graph's X-axis shows frequency in Hertz.
- The graph's Y-Axis shows the FFT of the Z-Axis accelerometer signal absolute value in metres per second squared (m/s2).

Magnetic field sensor

This section describes the information and specifications related to magnetic field sensor on the SMS.

This section describes the information and specifications related to magnetic field sensor on the SMS.

Measurement of the magnetic field around a motor allows for insights into electrical and mechanical motor performance issues.

ADI OtoSense™ SMS is equipped with a single magnetic field sensor For specifications, Table .

Table 5. Magnetic Field Sensor Specifications

Sensor Type	Sampling Frequency	Number of Samples	Data Formats	Data Type
Magnetic Field	6.25 kHz	15,000 samples over 2.4 seconds.	waveform, FFT	Float

Magnetic Field Time Waveform

This section describes the magnetic field time waveform. It is generated from data received by

This section describes the magnetic field time waveform. It is generated from data received by the ADI OtoSense™ Server from the magnetic field sensor on the SMS.

Access the Magnetic Field Time Waveform

This section describes the process of accessing the magnetic field time waveform graph.

This section describes the process of accessing the magnetic field time waveform graph.

To access the Magnetic Field Time Waveform through the application:

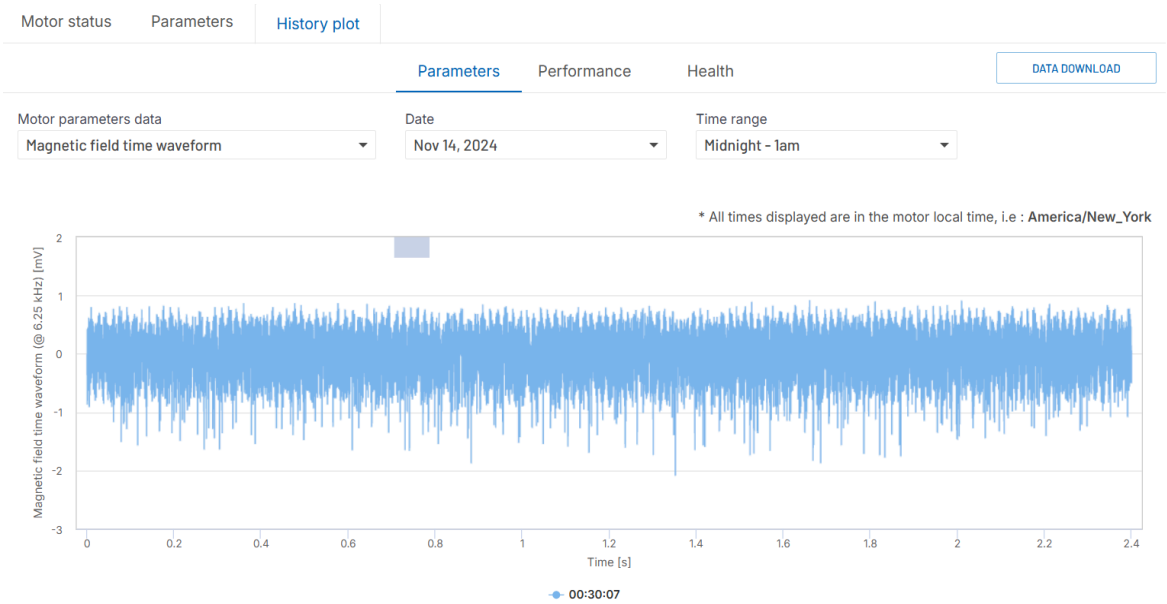
1. Tap **Motors**.

   The Locations list is displayed.

2. Select the location associated with the motor.
3. From the subsequent screens, select the corresponding process, plant, and motor name.
4. Select **History Plot** and **Parameters**.
5. Tap **Select Vibration** to select **Magnetic field time waveform**.

6. Tap **Select Date** and **Select Time** to select the date and time corresponding to the moment of the installation.

   You should see the corresponding waveform as shown in Figure 31 *(on page)*.



7. Click **DATA DOWNLOAD**to download the generated data.

Data Visualization

This section describes the magnetic field time waveform graph.

This section describes the magnetic field time waveform graph.

You can hover on a section of the magnetic field time waveform graph to view the magnetic field time waveform in millivolt (mV), the time in seconds, and the time frame.

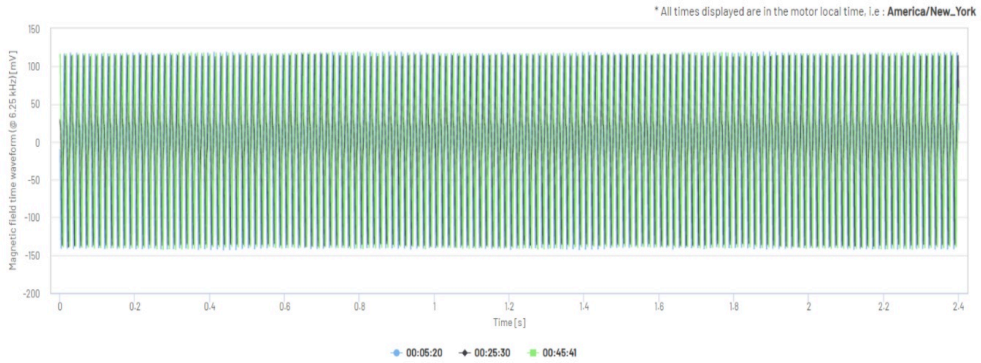
Figure 45. Image54



Sample Data

Three 2.4 second overlaid samples of the Magnetic field time waveform placeholder

- Figure 46. Three 2.4 second overlaid samples of the Magnetic field time waveform



- Figure 33 is a waveform graph showing three overlaid waveform samples from the Magnetic Field sensor.

- The graph's X-axis shows the time in seconds.
- The graph's Y-axis shows the time waveform in millivolt (mV) at at 6.25 kHz sample frequency.

Table 6. Magnetic field time waveform Legend

Legend	Description

Table 6. Magnetic field time waveform Legend (continued)

Figure 47. placeholder 	Waveform data at 5 minutes 20 seconds
Figure 48. placeholder 	Waveform data at 25 minutes 30 seconds
Figure 49. placeholder 	Waveform data at 45 minutes 41 seconds

- Figure 50. 2.4 seconds sample of the Magnetic field time waveform

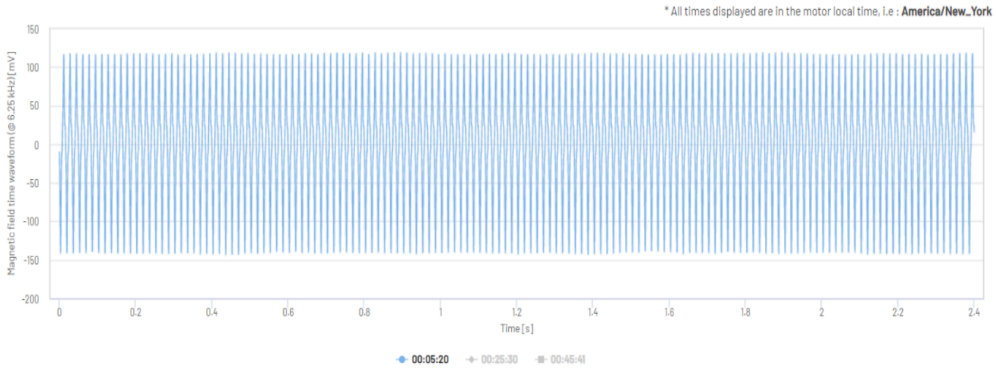
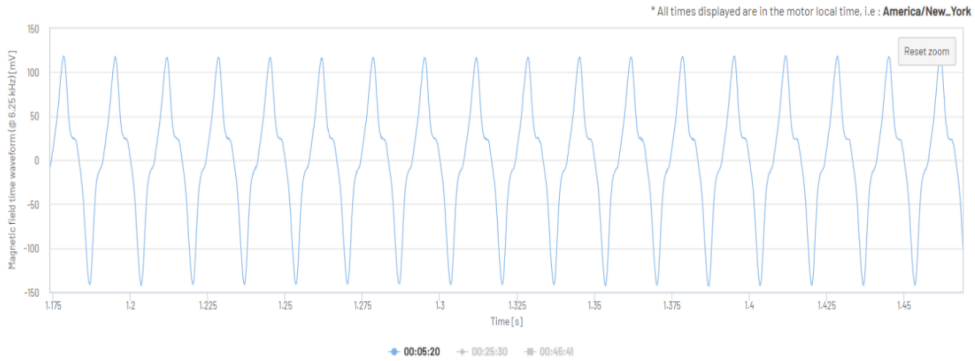


Figure 17 and shows the 2.4 second sample of the waveform data from the magnetic field sensor at 5 minutes 20 seconds (

).

- The graph's X-axis shows the time in seconds.
- The graph's Y-Axis shows the magnetic field waveform in millivolt (mV) at 6.25 kHz sample frequency.

- Figure 51. shows a zoomed in view of the same sample graph.



- Figure 35: Zoomed-in 2.4 seconds sample of the Magnetic field time waveform

API Dataset

This section describes the API dataset used for the magnetic field time waveform graph.

This section describes the API dataset used for the magnetic field time waveform graph.

The REST APIs have access to the following dataset:

flux

The Magnetic Field (flux) is the magnetic field leaked from the motor and measured on

The Magnetic Field (flux) is the magnetic field leaked from the motor and measured on the surface of the frame. This recorded data shows the magnetic field waveform data, in the unit of millivolt (mV). With the magnetic field from the motor, the analytics system can detect the electrical failures of the motor.

- Each record contains an array of 15,000 samples captured on the specified timestamp.

  It is necessary to generate the corresponding X-axis data to plot this Magnetic Field (flux) waveform data: The X-axis data is a vector of 15,000 data points with a step of 0.00016s.

- For example:

- [0, 0.00016, 0.00032, 0.00048, ... 2.39984]

- For more information on API refer to the ADI OtoSense™ SMS REST API.

Magnetic Field FFT dB

This section describes the magnetic field FFT. It is generated from data received by the

This section describes the magnetic field FFT. It is generated from data received by the ADI OtoSense™ Server from the magnetic field sensor on the SMS.

Access the Magnetic Field FFT dB

This section describes the process of accessing the magnetic field FFT graph.

This section describes the process of accessing the magnetic field FFT graph.

To access the Magnetic Field FFT dB through the application:

1. Tap **Motors**.

   The Locations list is displayed.

2. Select the location associated with the motor.

3. From the subsequent screens, select the corresponding process, plant, and motor name.

4. Select **History Plot** and **Parameters**.

5. Tap **Select Vibration** to select **Magnetic field FFT dB**.

6. Tap **Select Date** and **Select Time** to select the date and time corresponding to the moment of the installation.

   You should see the corresponding waveform as shown in Figure 36 *(on page)*.



7. Click **DATA DOWNLOAD**to download the generated data.

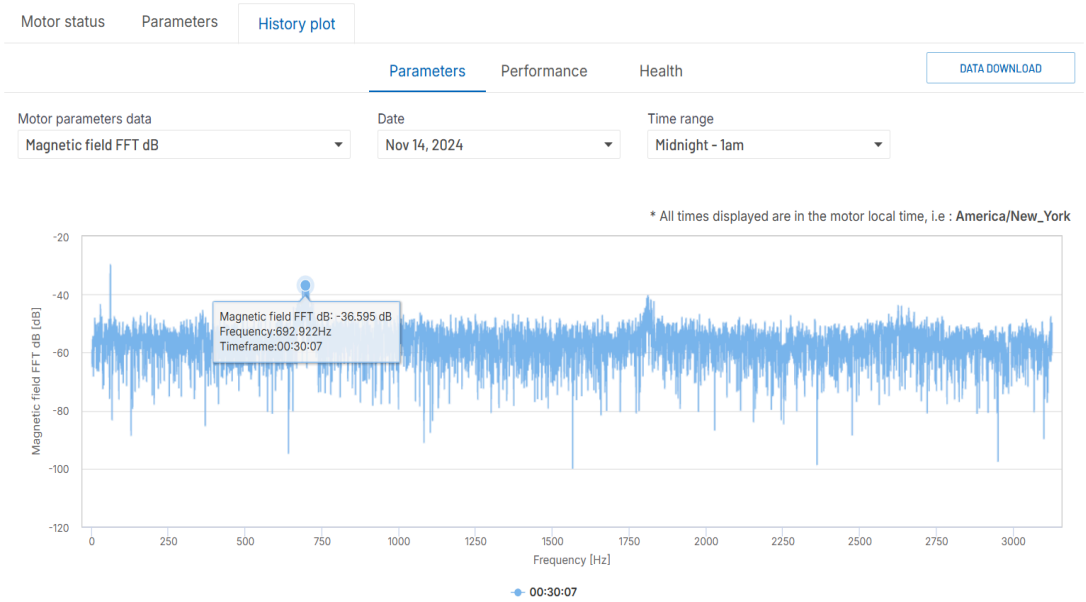
Data Visualization

This section describes the magnetic field FFT graph.

This section describes the magnetic field FFT graph.

You can hover on a section of the magnetic field FFT graph to view the magnetic field FFT dB in decibels (dB), the frequency in Hertz (Hz), and the time frame.

Figure 52. Image63



Sample Data

2.4 seconds sample of the Magnetic field FFT waveform placeholder

Figure 53. Image64

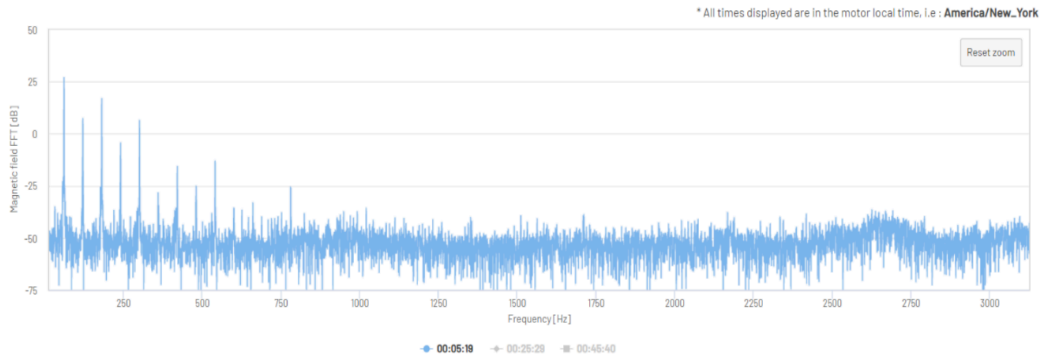


Figure 38 is a waveform graph showing Fast Fourier Transform in decibels data from the Magnetic Field sensor.

- The graph's X-axis shows frequency in Hertz.
- The graph's Y-Axis shows the FFT of the Magnetic Field in decibels.

API Dataset

This section describes the API dataset used for the magnetic field FFT graph.

This section describes the API dataset used for the magnetic field FFT graph.

The REST APIs have access to the following dataset:

fluxFFT

The Magnetic Field FFT (fluxFFT) is the magnetic field leaked from the motor and measured

The Magnetic Field FFT (fluxFFT) is the magnetic field leaked from the motor and measured on the surface of the frame. This recorded data shows the magnetic field Fast Fourier Transform data, in the unit of decibels (dB). With the magnetic field from the motor, the analytics system can detect the electrical failures of the motor.

- Each record contains an FFT of the flux samples captured on the specified timestamp.

  It is necessary to generate the corresponding X-axis data to plot this fluxFFT data in a graphic. The X-axis data is a vector of 7,500 data points with a step of 0.41667 Hertz (Hz).

- For example:

- [0.41667, 0.83334, 1.25, ... , 3125]

- For more information on API refer to the ADI OtoSense™ SMS REST API.

Magnetic Field FFT Absolute Value

This section describes the magnetic field FFT absolute value. It is generated from data received

This section describes the magnetic field FFT absolute value. It is generated from data received by the ADI OtoSense™ Server from the magnetic field sensor on the SMS.

Access the Magnetic Field FFT Absolute Value

This section describes the process of accessing the magnetic field FFT Absolute Value graph.

This section describes the process of accessing the magnetic field FFT Absolute Value graph.

To access the Magnetic Field FFT Absolute Value through the application:

1. Tap **Motors**.

   The Locations list is displayed.

2. Select the location associated with the motor.

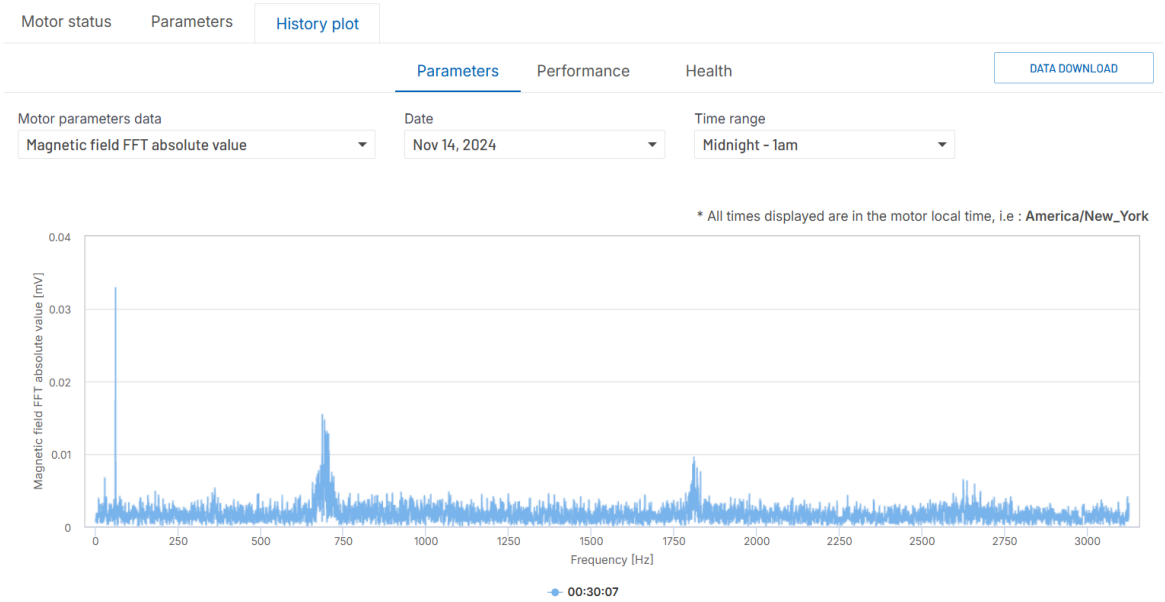
3. From the subsequent screens, select the corresponding process, plant, and motor name.

4. Select **History Plot** and **Parameters**.

5. Tap **Select Vibration** to select **Magnetic field FFT absolute value**.

6. Tap **Select Date** and **Select Time** to select the date and time corresponding to the moment of the installation.

   You should see the corresponding waveform as shown in Figure 39 *(on page)*.



7. Click **DATA DOWNLOAD**to download the generated data.

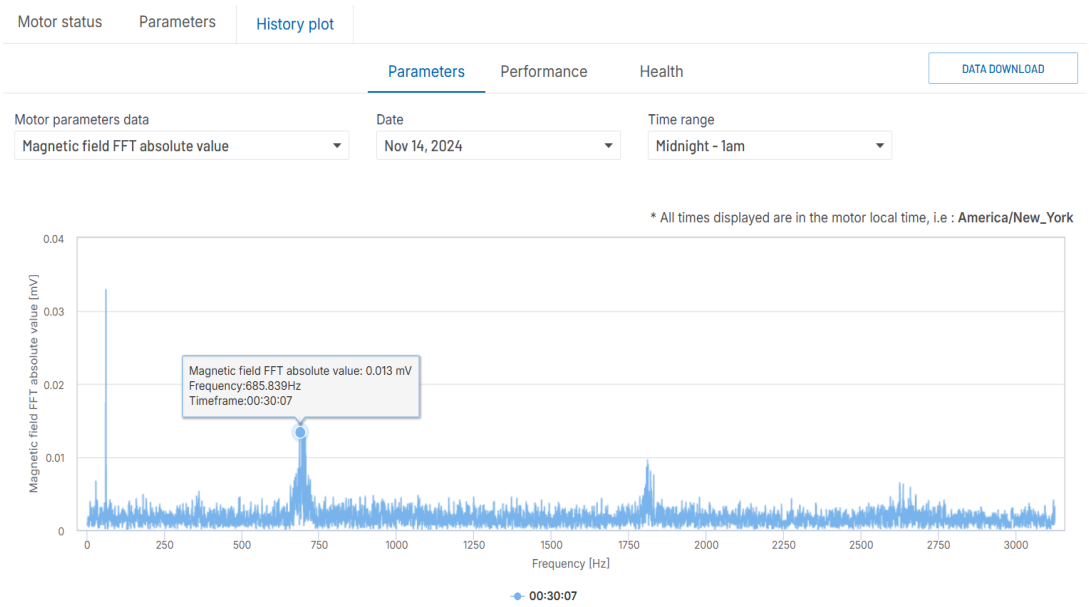
Data Visualization

This section describes the magnetic field FFT Absolute Value graph.

This section describes the magnetic field FFT Absolute Value graph.

You can hover on a section of the magnetic field FFT absolute value graph to view the magnetic field FFT absolute value in millivolt (mV), the frequency in Hertz (Hz), and the time frame.

Figure 54. Image66



Sample Data

Sample of the Magnetic field FFT absolute value waveform placeholder

Figure 55. Image67

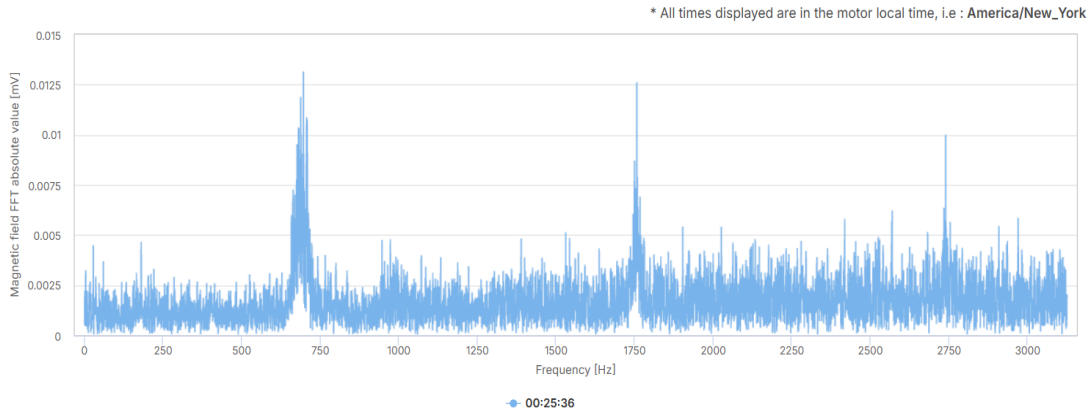


Figure 41 is a waveform graph showing Fast Fourier Transform absolute value in millivolts (mV) data from the magnetic field sensor.

- The graph's X-axis shows frequency in Hertz.
- The graph's Y-Axis shows the absolute value of the FFT of Magnetic Field in millivolts (mV).

Temperature sensors

This section describes the information and specifications related to the temperature sensors on the SMS.

This section describes the information and specifications related to the temperature sensors on the SMS.

Most motor faults will eventually result in high temperatures. These high temperatures can be an indication of the excessive friction within a motor or of OtoSense™ SMS is equipped with two on-board temperature sensors: an environment temperature sensor and a motor temperature sensor. For specifications, OBJ Table OBJ.

The environment temperature sensor allows for ADI OtoSense™ SMS to take ambient operating conditions, such as sunlight exposure, into account when assessing a motor's temperature status.

Table 7. Temperature Sensor Specifications

Sensor Type	Measurement Range	Precision Accuracy	Data Type
Environment Temperature			
Motor Temperature			

Environment Temperature

This section describes the environment temperature scatter plot. It is generated from data received by

This section describes the environment temperature scatter plot. It is generated from data received by the ADI OtoSense™ Server from the outer temperature sensor on the SMS.

Access Environment Temperature

This section describes the process of accessing the environment temperature scatter plot.

This section describes the process of accessing the environment temperature scatter plot.

To access the Environmental Temperature scatter plot through the application:

1. Tap **Motors**.

   The Locations list is displayed.

2. Select the location associated with the motor.

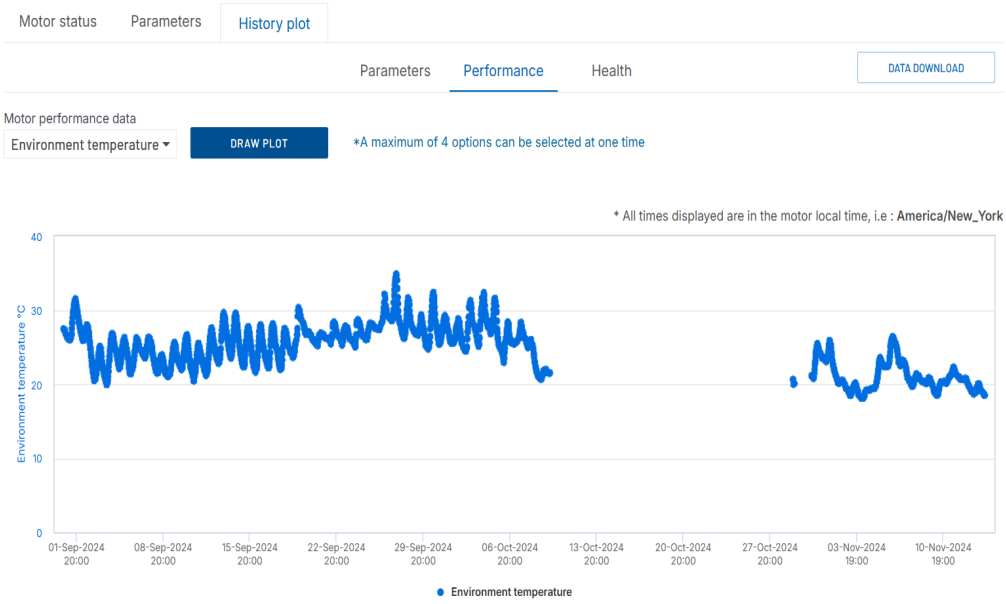
3. From the subsequent screens, select the corresponding process, plant, and motor name.

4. Select **History Plot** and **P erformance**.

5. Tap **Select …** to select **Environment temperature**.

   You should see the corresponding scatter plot as shown in Figure 42 *(on page)*.

Figure 56. Dashboard screen showing the Environment temperature scatter plot



6. Click **DATA DOWNLOAD**to download the generated data.

Data Visualization

This section describes the environment temperature scatter plot graph.

This section describes the environment temperature scatter plot graph.

You can hover on each data point (



) to view the environmental temperature, date, and time.

Figure 57. Dashboard screen highlighting one data point of the Environment temperature scatter plot



Sample Data

Sample of the Environment Temperature scatter plot placeholder

Figure 58. Sample of the Environment Temperature scatter plot

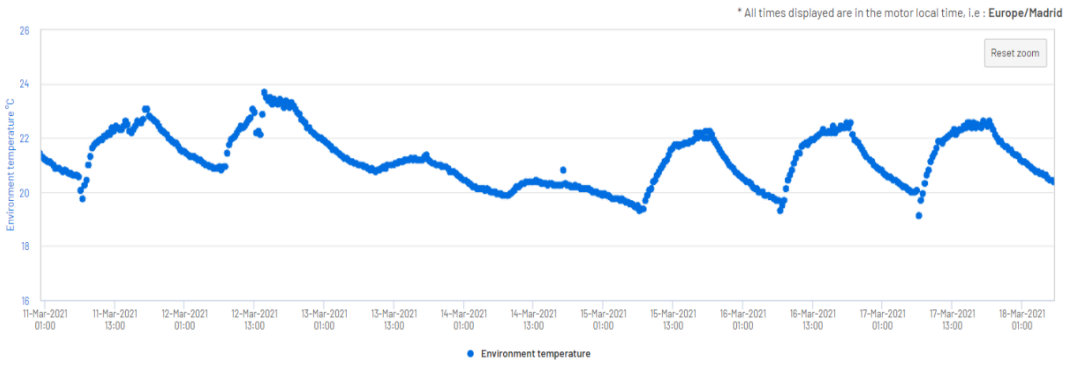


Figure 44 is a scatter plot showing the Environmental Temperature in Celsius (°C) from the Environmental Temperature sensor.

- The scatter plot's X-axis shows date and time.
- The scatter plot's Y-Axis shows the temperature in Celsius (°C).

API Dataset

This section describes the API dataset used for the environment temperature scatter plot.

This section describes the API dataset used for the environment temperature scatter plot.

The REST APIs have access to the following dataset:

tempe

The Environmental Temperature (tempe) is the temperature of the motor's surrounding area. There are two

The Environmental Temperature (tempe) is the temperature of the motor's surrounding area. There are two temperature sensors on both sides of the SMS: the temperature sensor on the outside side (not towards the motor) measures the environment temperature.

The recorded data shows the environmental temperature scatter plot data, in the unit of Celsius (°C). The analytics system can use this data to detect the electrical or mechanical failures of the motor.

Each record contains an array of two samples captured on the specified timestamp.

For more information on API refer to the ADI OtoSense™ SMS REST API.

Motor Temperature

This section describes the motor temperature scatter plot. It is generated from data received by

This section describes the motor temperature scatter plot. It is generated from data received by the ADI OtoSense™ Server from the inner temperature sensor on the SMS.

Access Motor Temperature

This section describes the process of accessing the motor temperature scatter plot.

This section describes the process of accessing the motor temperature scatter plot.

To access the Motor Temperature scatter plot through the application:

1. Tap **Motors**.

   The Locations list is displayed.

2. Select the location associated with the motor.

3. From the subsequent screens, select the corresponding process, plant, and motor name.

4. Select **History Plot** and **P erformance**.

5. Tap **Select …** to select **Motor temperature**.

   You should see the corresponding scatter plot as shown in Figure 45 *(on page)*.

Figure 59. Dashboard screen showing the Motor temperature scatter plot



6. Click **DATA DOWNLOAD**to download the generated data.

Data Visualization

This section describes the motor temperature scatter plot graph.

This section describes the motor temperature scatter plot graph.

You can hover on each data point (



) to view the motor temperature, date, and time.

Figure 60. Dashboard screen highlighting one data point of the Motor temperature scatter plot



Sample Data

Sample of the Motor Temperature scatter plot placeholder

Figure 61. Sample of the Motor Temperature scatter plot

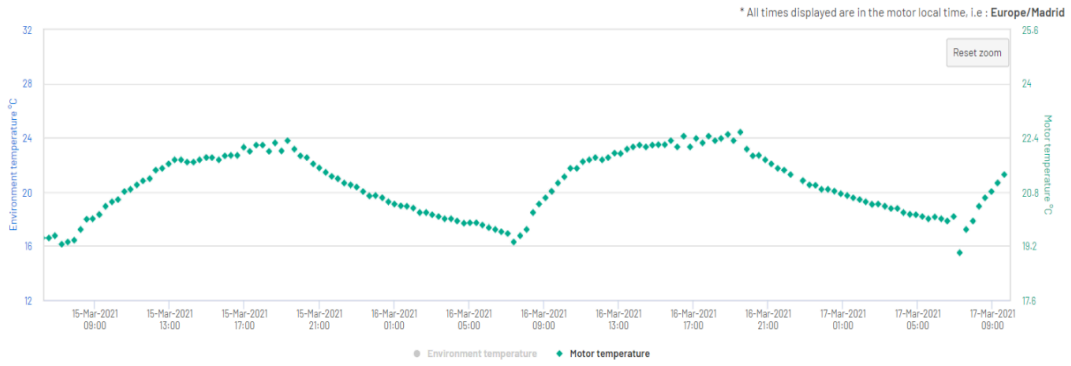


Figure 47 is a scatter plot showing the Motor Temperature in Celsius (°C) from the Motor Temperature sensor.

- The scatter plot's X-axis shows date and time.
- The scatter plot's Y-Axis shows the temperature in Celsius (°C).

API Dataset

This section describes the API dataset used for the motor temperature scatter plot.

This section describes the API dataset used for the motor temperature scatter plot.

The REST APIs have access to the following dataset:

tempm

The Motor Temperature (tempm) is the motor frame temperature measured by the SMS. The temperature

The Motor Temperature (tempm) is the motor frame temperature measured by the SMS. The temperature sensor on the inside side (towards the motor) measures the motor temperature.

The recorded data shows the motor temperature scatter plot data, in the unit of Celsius (°C). The analytics system can use this data to detect the electrical or mechanical failures.

Each record contains an array of two samples captured on the specified timestamp.

For more information on API refer to the ADI OtoSense™ SMS REST API.

Chapter 5. Motor Object

This section describes the motor object.

This section describes the motor object.

Overview

The motor object consists of a components map. Each component has a Fault Score Index

The motor object consists of a components map. Each component has a Fault Score Index (FSI) and a Code. The FSI is a value between 0 and 10, which indicates the following motor conditions or state:

- FSI < ‘5’: **Alarm**
- FSI= ‘5’ or ‘6’: **Warning**
- FSI > ‘6’: **Good**.

Note:

Note: The lower the FSI value the greater the severity of the component fault.

For each component in a **Warning** or an **Alarm** state, an appropriate **Code** is assigned by the ADI OtoSense™ Cloud analytics. Each code defines both the recommended Action and the Description of the associated fault.

Each component has multiple possible codes depending on the severity of the fault.

For example, a Power System component with an FSI of ‘3’ or ‘4’ will be assigned code ‘1002’ and a Power System component with an FSI of ‘2’ or below will be assigned code ‘1003’. For more information, see Table 10.

Component Code

This section describes the component code of the motor object.

This section describes the component code of the motor object.

Table 8. Component Codes Definitions

Code	Action	Description	Notes

Table 8. Component Codes Definitions (continued)

-1	Check SMS	Disconnected	
0	No action required	No fault	
1001	Check this condition over time	Warning: possible power system imbalance	
1002	Action Required: check motor power system	Alarm: power system imbalance detected	
1003	Action Required: stop motor and check motor power system	Alarm: high power system imbalance detected	
2001	Check this condition over time	Warning: possible stator short-circuit	
2002	Action Required: check stator winding condition	Alarm: stator short-circuit detected	
2003	Action Required: stop motor and check stator winding condition	Alarm: stator short-circuit detected	
3001	Check this condition over time	Warning: possible rotor active parts problem	
3002	Action Required: check rotor condition	Alarm: rotor active parts problem detected	
3003	Action Required: replace rotor	Alarm: rotor active parts problem detected	
4001	Check this condition over time	Warning: imbalance detected	
4002	Action Required: balance motor in next stop	Alarm: imbalance detected	
4003	Action Required: balance motor as soon as possible	Alarm: high imbalance detected	
5001	Check this condition over time	Warning: possible eccentricity	
5002	Action Required: check rotor eccentricity	Alarm: eccentricity detected	
5003	Action Required: check rotor eccentricity	Alarm: serious eccentricity detected	
6001	Check this condition over time	Warning: possible bearing outer race malfunction	

Table 8. Component Codes Definitions (continued)

6002	Action Required: lubricate bearings	Alarm: bearing outer race fault detected	
6003	Action Required: replace bearings as soon as possible	Alarm: bearing outer race high fault detected	
6004	Check this condition over time	Warning: possible bearing inner race malfunction	
6005	Action Required: lubricate bearings	Alarm: bearing inner race fault detected	
6006	Action Required: replace bearings as soon as possible	Alarm: bearing inner race high fault detected	
6007	Check this condition over time	Warning: possible bearing rolling element malfunction	
6008	Action Required: lubricate bearings	Alarm: bearing rolling element fault detected	
6009	Action Required: replace bearings as soon as possible	Alarm: bearing rolling element high fault detected	
6010	Check this condition over time	Warning: possible bearing cage malfunction	
6011	Action Required: lubricate bearings	Alarm: bearing cage fault detected	
6012	Action Required: replace bearings as soon as possible	Alarm: bearing cage high fault detected	
6013	Check this condition over time	Warning: possible DE (Drive End) bearing outer race malfunction	
6014	Action Required: lubricate bearings	Alarm: DE bearing outer race fault detected	
6015	Action Required: replace bearings as soon as possible	Alarm: DE bearing outer race high fault detected	
6016	Check this condition over time	Warning: possible DE bearing inner race malfunction	
6017	Action Required: lubricate bearings	Alarm: DE bearing inner race fault detected	

Table 8. Component Codes Definitions (continued)

6018	Action Required: replace bearings as soon as possible	Alarm: DE bearing inner race high fault detected	
6019	Check this condition over time	Warning: possible DE bearing rolling element malfunction	
6020	Action Required: lubricate bearings	Alarm: DE bearing rolling element fault detected	
6021	Action Required: replace bearings as soon as possible	Alarm: DE bearing rolling element high fault detected	
6022	Check this condition over time	Warning: possible DE bearing cage malfunction	
6023	Action Required: lubricate bearings	Alarm: DE bearing cage fault detected	
6024	Action Required: replace bearings as soon as possible	Alarm: DE bearing cage high fault detected	
6025	Check this condition over time	Warning: possible NDE (Non-Drive End) bearing outer race malfunction	
6026	Action Required: lubricate bearings	Alarm: NDE bearing outer race fault detected	
6027	Action Required: replace bearings as soon as possible	Alarm: NDE bearing outer race high fault detected	
6028	Check this condition over time	Warning: possible NDE bearing inner race malfunction	
6029	Action Required: lubricate bearings	Alarm: NDE bearing inner race fault detected	
6030	Action Required: replace bearings as soon as possible	Alarm: NDE bearing inner race high fault detected	
6031	Check this condition over time	Warning: possible NDE bearing rolling element malfunction	
6032	Action Required: lubricate bearings	Alarm: NDE bearing rolling element fault detected	

Table 8. Component Codes Definitions (continued)

6033	Action Required: replace bearings as soon as possible	Alarm: NDE bearing rolling element high fault detected	
6034	Check this condition over time	Warning: possible NDE bearing cage malfunction	
6035	Action Required: lubricate bearings	Alarm: NDE bearing cage fault detected	
6036	Action Required: replace bearings as soon as possible	Alarm: NDE bearing cage high fault detected	
7001	Check this condition over time	Warning: possible misalignment	
7002	Action Required: align motor in next stop	Alarm: misalignment detected	
7003	Action Required: align motor as soon as possible	Alarm: high misalignment detected	
8001	Check this condition over time	Warning: cooling system malfunction	
8002	Action Required: check cooling system	Alarm: cooling system malfunction	
8003	Action Required: check cooling system as soon as possible	Alarm: dangerous cooling system malfunction	
9001	Check this condition over time	Warning: possible loose foot – motor frame problem	
9002	Action Required: check motor-bench assembly	Alarm: loose foot - motor frame problem detected	
9003	Action Required: check motor-bench assembly	Alarm: loose foot - motor frame severe problem detected	
10001	Check this condition over time	global vibration acceleration increasing <50%	
10002	Check this condition over time	global vibration acceleration increasing <70%	
10003	Stop Motor: check bearing lubrication, bearings condition, coupling and mechanical parts	global vibration acceleration increasing <100%	
10004	Stop Motor: check bearing lubrication, bearings condition, coupling and mechanical parts	global vibration acceleration increasing <200%	

Table 8. Component Codes Definitions (continued)

10005	Stop Motor: check bearing lubrication, bearings condition, coupling and mechanical parts	global vibration acceleration increasing <250%	
10006	Stop Motor: check bearing lubrication, bearings condition, coupling and mechanical parts	global vibration acceleration increasing <300%	
10007	Stop Motor: check bearing lubrication, bearings condition, coupling and mechanical parts	global vibration acceleration increasing >300%	
10008	Check this condition over time	global vibration velocity increasing <50%	
10009	Check this condition over time	global vibration velocity increasing <70%	
10010	Stop Motor: coupling alignment, structure looseness, bearings condition and mechanical parts	global vibration velocity increasing <100%	
10011	Stop Motor: coupling alignment, structure looseness, bearings condition and mechanical parts	global vibration velocity increasing <200%	
10012	Stop Motor: coupling alignment, structure looseness, bearings condition and mechanical parts	global vibration velocity increasing <250%	
10013	Stop Motor: coupling alignment, structure looseness, bearings condition and mechanical parts	global vibration velocity increasing <300%	
10014	Stop Motor: coupling alignment, structure looseness, bearings condition and mechanical parts	global vibration velocity increasing <300%	
10015	Check this condition over time	Warning: motor's RPM has increased	
10016	Action Required: if this is normal operation mode for the motor, please perform relearning	Alarm: motor's RPM is higher than normal operation	
10017	Action Required: if this is normal operation mode for the motor, please perform relearning	Alarm: motor's RPM is much higher than normal operation	
10018	Check this condition over time	Warning: motor's RPM has decreased	
10019	Action Required: if this is normal operation mode for the motor, please perform relearning	Alarm: motor's RPM is lower than normal operation	
10020	Action Required: if this is normal operation mode for the motor, please perform relearning	Alarm: motor's RPM is much lower than normal operation	

Table 8. Component Codes Definitions (continued)

10021	Check this condition over time	Warning: motor's load has increased	
10022	Action Required: if this is normal operation mode for the motor, please perform relearning	Alarm: motor's load is higher than normal operation	
10023	Action Required: if this is normal operation mode for the motor, please perform relearning	Alarm: motor's load is much higher than normal operation	
10024	Check this condition over time	Warning: motor's load has decreased	
10025	Action Required: if this is normal operation mode for the motor, please perform relearning	Alarm: motor's load is lower than normal operation	
10026	Action Required: if this is normal operation mode for the motor, please perform relearning	Alarm: motor's load is much lower than normal operation	
10027	Check this condition over time	Warning: the environment temperature has increased	
10028	Action Required: if this is normal operation mode for the motor, please perform relearning	Alarm: the environment temperature is higher than normal operation	
10029	Action Required: if this is normal operation mode for the motor, please perform relearning	Alarm: the environment temperature is much higher than normal operation	
10030	Check this condition over time	Warning: the environment temperature has decreased	
10031	Action Required: if this is normal operation mode for the motor, please perform relearning	Alarm: the environment temperature is lower than normal operation	
10032	Action Required: if this is normal operation mode for the motor, please perform relearning	Alarm: the environment temperature is much lower than normal operation	

Motor Log Events

This section describes the events of the motor log.

This section describes the events of the motor log.

Each Motor has an associated Motor Log. This log has historical entries that record the various events that the motor has experienced:

Table 9. Motor Log Events

State	Description
`learning`	Motor is learning to generate model
`alarm`	One or more components have an FSI below ‘5’
`comment`	Operator added a comment through the cloud web or mobile application
`repaired`	Operator took remedial action and marked the motor as repaired
`recovered`	Motor condition has improved; the FSI has increased to ‘5’ or above

Chapter 6. Appendices

Appendix A: Miscellaneous API Datasets

This section describes the miscellaneous data sets.

This section describes the miscellaneous data sets.

performance

This section describes the parameters used to assess motor performance.

This section describes the parameters used to assess motor performance.

Each record contains a collection of performance parameters:

Table 10. Performance Parameters

Parameter	Measurement Value	Description
`rmsVx`	m/s2	RMS value for the vibration X acceleration (tangential)
`rmsVxV`	mm/s	RMS value for the vibration X velocity (tangential)
`rmsVxD`	µm	RMS value for the vibration X displacement (tangential)
`rmsVz`	m/s2	RMS value for the vibration Z acceleration (axial)
`rmsVzV`	mm/s	RMS value for the vibration Z velocity (axial)

Table 10. Performance Parameters (continued)

`rmsVzD`	µm	RMS value for the vibration Z displacement (axial)
`rmsFlux`	mV	RMS value for the magnetic field
`rpm`	RPM (rev/min)	Motor rotating speed in RPM
`tempMotor`	ºC	Temperature of the motor (ºC)
`tempEnv`	ºC	Temperature of the motor environment (ºC)
`operationHours`	hours	Approximation of the motor running time
`running`	Integer flag	Running flag 1 if running 0 if not running When the motor is rotating at a very low speed, less than 25% of its rated speed (rated RPM), the analytics may interpret the motor as stopped. `running is used to calculate the Number of Starts in a time range. Chronologically sort the performance records in the time range and count the number of times _running_ transitions from '0' to '1'. This approximates the number of times the motor started in the specified time range.`

For more information on API refer to the ADI OtoSense™ SMS REST API.

conditions

This section describes the parameters used to assess the condition of a motor component.

This section describes the parameters used to assess the condition of a motor component.

Each parameter in the following table represents the Fault Score Index (FSI) of the corresponding motor component condition. These predicted health indicators for motor components are generated each time the analytics processes data samples received from a running motor. The values are between zero and ten:

- A value of 10 corresponds to a perfect state for the component.
- A value of 0 corresponds to a catastrophic state for the component.

Table 11. Condition Parameters

Parameter	Motor Component	Description
`powerSystem`	Power System	Problems in the three phases of the power supply, which could lead to a motor current imbalance.
`statorWinding`	Stator Winding	Problems in one of the phases of the motor, which could cause an imbalance of the motor currents.
`rotorBar`	Rotor	Short-circuit ring or rotor bar related problems.
`imbalance`	Motor Shaft/ Balance	Unequal distribution of mass, causing the center of gravity to shift from the center of rotation.
`eccent`	Eccentricity	Asymmetric air gap between the rotor and stator.

Table 11. Condition Parameters (continued)

`ricity`		
`bearing`	Bearing	Mechanical stress or contamination leading to small cracks or defects that occur in the bearing, creating vibration problems.
`misalignment`	Alignment	Occurs when two rotating shafts (motor and load) are not aligned, creating external misalignment.
`cooling`	Cooling System	Problems with fans attached to the shaft or externally attached to the motor.
`motorStructure`	Loose Foot	Structural looseness occurring when the motor base (or connection to the motor base) is not properly tightened.
`performance`	Performance	Used to capture abnormal events such as an overall increase in vibrations, unusual changes in motor RPM, motor load or temperatures, etc. Indicates faults that do not correspond to any of the other nine fault categories, which could be related to machine process.

For more information on API refer to the ADI OtoSense™ SMS REST API.

operations

This section describes the parameter used to assess the battery status of the SMS.

This section describes the parameter used to assess the battery status of the SMS.

Each record contains operational details captured on the specified timestamp.

Table 12. Operations Parameters

Parameter	Description
`batteryStatus`	Battery status, g for full, y for medium, r for low.

For more information on API refer to the ADI OtoSense™ SMS REST API.

Appendix B: Cloud Platform Parameters

This section describes the cloud platform parameters.

This section describes the cloud platform parameters.

Table 13. Cloud Platform Parameters

Parameter	Description	Data Format	Unit
`vibx`	Vibration X-axis waveform	array	m/s2
`vibz`	Vibration Z-axis waveform	array	m/s2
`flux`	Magnetic flux	array	mV
`vibxFFT`	Vibration X-axis FFT magnitude	array	dB
`vibzFFT`	Vibration Z-axis FFT magnitude	array	dB
`fluxFFT`	Magnetic flux FFT magnitude	array	dB
`tempe`	Environment temperature	array	°C
`tempm`	Motor temperature	array	°C
`vbat`	Battery voltage	array	V

Appendix C: ADI OtoSense™ SMS Technical Specifications

This section describes the technical specifications of the SMS.

This section describes the technical specifications of the SMS.

Table 14. SMS Technical Specifications

Specification	Value
Networking Standards	Wi-Fi b/g/n
Radio Standards	IEEE 802.11 b/g/n
Networking Frequency	2.4 GHz
Wi-Fi Security Standards	WPA2/3
Wireless Range	>50 m
Battery Life	1 year (estimated)
Operating Temperature	-40°C to +60°C
Storage Temperature	50°C max
Weight	0.5 Kg
Certifications	 IP Class IP55

FAQs

To view frequently asked questions about the SMS device, the installation process, and the web

To view frequently asked questions about the SMS device, the installation process, and the web and mobile applications, see FAQs

Documents

To access the ADI OtoSense™ SMS Installation and Web User Guides, see Guides

To access the ADI OtoSense™ SMS Installation and Web User Guides, see Guides

Index

A

absolute
31, 32, 41, 42, 51, 52

Absolute
30, 30, 40, 40, 50, 50

access
73

Action
60

Alarm
60

amplitude
40

analytics
4

Appendix
68

application
6

array
72

asked
73

attribute
17

Audience
4

B

based
5

battery
71

benefits
5

Benefits
5

C

Cloud
72

component
60, 69

D

Data
20, 24, 28, 29, 34, 38, 39, 49, 54, 57

dataset
26, 29, 36, 40, 47, 50, 55, 58

Dataset
26, 29, 36, 40, 47, 50, 55, 58

datasets
26

Datasets
68

describes
27, 37

Documents
73

E

Each
60

Environment
53, 53

Events
66

F

FAQs
73

fault
60

Fault
12

field
22, 43, 44, 47, 49, 50, 51
Field
43, 47, 48, 50
Foot
18
frequently
73
Fully
11

G

generate
30
graph
29, 35, 49
guide
4
Guides
73

H

health
17
Health
18, 18, 18, 19, 20
Historical
19

I

Index
18, 19, 20
Information
22
Introduction
5

L

learning
8, 9
Learning
8, 9, 11
Loose
18

M

magnetic
44, 47, 47, 49, 50, 51
Magnetic
43, 43, 43, 47, 48, 49, 50, 50
Miscellaneous
68
mobile
6
Mode
11
Modes
8
motor
5, 5, 5, 8, 8, 12, 16, 17, 36, 47, 50, 56, 59, 66, 68, 69
Motor
16, 56, 56, 58, 60, 66

N

Normal
11

O

Object
60
Operating
8
Operation
11
operations
71
OtoSense
5, 8
Overview
22

P

Performance
18
placeholder
6, 12, 25, 45
Predictive
11, 11

Q

Quick
9

R

Required
60

S

Sample
29, 32, 39, 42, 49, 52, 55, 58
scatter
53, 55, 56, 58
seconds
25, 35, 45
section
8, 26, 27, 29, 36, 37, 40, 47, 50, 55, 58, 60
select
27, 37
Select
23, 27, 30, 33, 37, 40, 43, 48, 50
sensor
43
Sensor
22
sensors
22, 22, 52
Server
8
shows
39
Specifications
22, 73
stage
11, 11
Standards
73
Stator
18
status
16, 71
system
5
System
69

T

Technical
73
tempe
56
temperature
52, 53, 54, 56, 56, 57, 59
Temperature
52, 53, 53, 55, 56, 56
tempm
59
Time
23, 23, 33, 33, 43, 43

V

value
31, 32, 41, 42, 52, 68
Value
30, 40
vibration
22, 26, 30, 36, 40, 68
Vibration
22, 23, 23, 24, 27, 27, 28, 30, 30, 30, 31, 33, 33, 34, 36, 37, 37, 38, 40, 40, 40, 41, 72
Visualization

28, 38, 54, 57

W

waveform
24, 25, 26, 34, 35, 44, 45
Waveform
23, 33
Winding
18